

RESEARCH ARTICLE

A Reinforcement Learning Approach Combined With Scope Loss Function for Crime Prediction on Twitter (X)

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This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the Social Sciences Ethics Committee of the University of Chicago.

ABSTRACT Online social networks, especially Twitter (X), have become focal points for illicit activities, providing unique criminal investigation opportunities. This paper introduces an innovative methodology that uses social media sentiment analysis to predict criminal activities. One major challenge in sentiment analysis is the uneven distribution of sentiment classes, where traditional models often fail to accurately classify instances of the minority class due to the overwhelming presence of majority class data. To tackle this issue, we propose a model that combines a reinforcement learning (RL) algorithm with a scope loss function. The RL approach uses a reward mechanism that assigns a more significant value to correctly predicting minority class instances over majority class ones. The scope loss function ensures an optimal balance between utilizing known data and exploring new data, thus maintaining a delicate equilibrium between accuracy and generalizability. Our model employs a series of convolutional neural networks (CNNs) to extract significant features from textual content, which are then utilized for sentiment classification. We also incorporate an advanced artificial bee colony (ABC) optimization technique to refine the model's hyperparameters. The effectiveness of our approach was empirically tested using two distinct datasets: one consisting of crime incident reports from the Chicago Police Department covering the period from September 2019 to July 2024 and another comprising tweets containing crime-related terms related to Chicago. The predictive capabilities of our proposed model were benchmarked against existing models, demonstrating superior performance with accuracies of 96.411% and 94.088%, respectively. This breakthrough highlights the potential of integrating sentiment analysis with reinforcement learning to significantly enhance the predictive accuracy of crime-related activities in online social networks, offering valuable insights for law enforcement and criminal investigation applications.

INDEX TERMS Crime prediction, sentiment analysis, reinforcement learning, class imbalance, hyperparameter tuning, artificial bee colony.

I. INTRODUCTION

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Social media, especially platforms like Twitter (X) with its 300 million users, has woven itself into the fabric of daily

life, offering a space where people can share texts, images, and sounds and form connections beyond superficial friendships [1]. This creates a rich tapestry of public discussions and personal expressions embedded in vast textual data. This data, reflecting users' daily activities and interactions, holds untapped potential for predicting criminal activities. However, the challenge lies in the ever-evolving nature of crime, which continuously adapts and morphs, making static prediction models quickly outdated. Artificial Intelligence (AI) emerges as a game-changer in this context, significantly boosting the capability to predict and detect crimes on platforms like Twitter (X). Leveraging advanced AI techniques, including machine learning and natural language processing, allows for the real-time analysis of the extensive, unstructured data generated by social media. This analysis can reveal hidden patterns, anomalies, and emerging trends indicative of potential criminal behavior or threats. AI's ability to learn and adapt to new patterns of crime as they develop ensures that predictive models remain relevant and practical, even as criminals change their tactics [2].

Using sentiment analysis for crime prediction has recently become a promising strategy [3], [4], [5], [6]. Sentiment analysis uses text and computational linguistics to understand and categorize people's opinions in unstructured text [7]. Its main aim is to analyze the tone of sentences, identifying whether they are positive, negative, or neutral, by focusing on the context of words [8]. This technique helps draw insights from unstructured data like tweets or online reviews [9]. However, one major hurdle in sentiment analysis is the imbalance between negative (non-crime-related) and positive (crime-related) data samples. This imbalance can significantly lower the effectiveness of predictive models. To address this, various strategies have been tried, including methods to balance the dataset, like oversampling the less frequent samples or undersampling the more frequent ones, and more sophisticated approaches like the Synthetic Minority Over-sampling Technique (SMOTE) [10], [11]. However, these methods can lead to loss of information or cause the model to overfit the training data [12]. As an alternative, cost-sensitive learning approaches have been introduced [13]. These approaches assign different costs to misclassifying other classes, aiming to reduce the impact of imbalance. However, setting up a suitable cost matrix is tricky because the costs are only sometimes straightforward [14].

Deep learning-based methods have been proposed for classifying imbalanced data [15], [16], [17], [18], but they face several inherent challenges [19], [20]. One major issue is their heavy reliance on large, balanced datasets to achieve high performance, often different in real-world scenarios like crime prediction, where positive samples are far fewer than negative ones. This imbalance can lead to biased models that favor the majority class, resulting in poor predictive performance for the minority class. Furthermore, deep learning models can be opaque, making it difficult to understand how they make decisions, which is crucial in sensitive applications like crime

detection [21]. Deep reinforcement learning (DRL) offers a promising alternative by addressing these challenges through its reward system. Unlike traditional deep learning methods that learn from a fixed dataset, DRL agents learn by interacting with their environment, making decisions, and receiving feedback in the form of rewards [22]. This dynamic learning process allows DRL agents to focus on long-term outcomes and adjust their strategies accordingly. In the context of imbalanced datasets, a DRL agent can be designed to receive higher rewards for correctly identifying rare positive samples, incentivizing the model to pay more attention to the minority class [23]. This approach naturally balances the importance of different classes without the need for oversampling or undersampling techniques. Moreover, the reward system in DRL can be tailored to penalize the model for wrong predictions, especially for the minority class, further refining the model's ability to distinguish between classes accurately [24]. This flexibility allows DRL to adaptively learn from imbalanced data, potentially overcoming the limitations of traditional deep learning methods in applications like crime prediction, where accurately detecting the minority class is crucial [25].

In DRL, finding the right balance between using what is already known (exploitation) and discovering new possibilities (exploration) is crucial [26], [27]. This balance is critical because leaning too much towards exploitation can lead the RL agent to suboptimal decisions. When an agent over-focuses on exploitation, it relies solely on its existing knowledge, neglecting to explore other areas that might hold valuable insights [28]. This behavior can prevent the agent from encountering new and potentially beneficial states and actions, limiting its learning potential. To overcome this challenge, significant advancements have been made in the field. Researchers have developed sophisticated strategies, including advanced loss functions, to tackle the issue of data imbalance during the training of RL agents [29], [30]. These innovative approaches aim to correct skewed data distribution, ensuring the learning process is more balanced and effective. Providing a more equitable learning environment, these solutions help RL agents navigate the exploration-exploitation trade-off better, enhancing their overall performance and decision-making efficiency [31], [32].

Choosing the correct hyperparameters is also crucial for improving the effectiveness of DRL models and avoiding issues like overfitting and underfitting [33], [34]. Hyperparameters, which are settings determined before the model training begins, significantly influence how well the model can predict outcomes [35]. However, finding the best set of hyperparameters remains challenging, as there is no one-size-fits-all solution across different applications. Traditionally, methods like grid search [36] and genetic algorithms [37] have been employed for hyperparameter optimization. Grid search, though straightforward, is only practical for models with a limited number of parameters due to its exhaustive nature [38]. On the other hand, genetic algorithms offer more

flexibility but can be constrained by the need to categorize hyperparameters into discrete options [39]. This requirement can lead to less-than-ideal exploration of possible settings, particularly for parameters better tuned continuously [40]. The ABC algorithm [41] offers an innovative alternative that overcomes these limitations [42]. Unlike traditional methods, the ABC algorithm does not require discretizing hyperparameters, allowing for a more fluid and effective search process. It adeptly balances between focusing on promising areas (local exploration) and scouting out new possibilities (global exploration). This adaptability makes the ABC algorithm especially suited for DRL tasks that involve complex and vast hyperparameter spaces. Furthermore, its nature-inspired strategy is more systematic and computationally efficient than exhaustive approaches like grid search, making it an excellent tool for finding the best hyperparameter settings while keeping resource use in check [43].

This study addresses two significant challenges in using social media analysis for crime prediction: the problem of unbalanced classification in sentiment analysis and the complexities of hyperparameter optimization. We present a sentiment analysis model that employs three dilated convolution layers to extract essential feature vectors for classification tasks. To tackle the issue of unbalanced classification, we incorporate an RL algorithm with a scope loss function. This loss function adeptly manages the balance between exploiting known data and exploring new possibilities, thereby significantly enhancing the learning outcomes of our RL models. Our innovative approach combats performance degradation and adjusts for dataset imbalances by treating sentiment classification as a sequence of decision-making steps. In this model, an agent is rewarded for accurate classifications, with reduced rewards for identifying the more frequently occurring class. This method emphasizes the importance of recognizing underrepresented categories. Additionally, our model utilizes DistilBERT word embeddings, which streamline the original BERT model by reducing its size by 40%, thereby boosting computational efficiency. Furthermore, we employ an advanced ABC method for hyperparameter optimization, enhancing the model's overall effectiveness.

The effectiveness of our model was confirmed using two different datasets: crime incident reports from the Chicago Police Department and tweets that included terms related to crime in Chicago. Compared with existing models, ours demonstrated superior performance, achieving 96.411% and 94.088% accuracy rates, respectively. We conducted ablation studies to determine the contributions of the RL, scope loss function, and the hyperparameter optimization components to the model's performance. Additional experiments evaluated the influence of the proposed reward function, various loss functions, and different word embedding techniques on the model's overall efficacy.

The proposed model's key contributions include:

- Our model introduces a hybrid approach that leverages RL to tackle the persistent issue of imbalanced

classification in sentiment analysis. This method enables the model to dynamically adjust its focus between the more and less frequent categories, directly addressing the challenge where particular sentiments or outcomes are overrepresented. The RL component manipulates strategies based on real-time feedback, thus refining its predictive accuracy over time and enhancing the model's ability to deal with diverse data distributions.

- We have integrated a scope loss function into the RL framework to intricately balance the exploitation of known data and the exploration of new possibilities. This loss function is meticulously designed to foster an optimal learning environment that prevents the model from overfitting to prevalent data patterns while avoiding undue emphasis on less relevant features. By maintaining this balance, the scope loss function significantly improves the model's generalization capabilities across varied datasets.
- The model incorporates a unique reward system that not only effectively addresses data imbalance but also promotes fairness and balance in the classification process. This approach provides greater incentives for accurately classifying underrepresented categories, ensuring that minority classes are given due importance. By modifying the reward structure, our model encourages deeper learning about less-represented data points, which is crucial for comprehensive sentiment analysis.
- Our implementation of an ABC-based strategy for hyperparameter optimization brings significant enhancements to the model's robustness and accuracy. This strategy systematically searches for the optimal set of parameters, enabling the model to perform efficiently across various scenarios. The ABC method optimizes the critical parameters that influence learning, such as learning rate and regularization terms, thereby fine-tuning the model to achieve superior performance metrics.

The remainder of the article is structured as follows: Section II reviews relevant literature on crime prediction, Section III outlines preliminary and background information, Section IV details the proposed methodology, Section V presents experimental results and analyses, and Section VI concludes the article.

II. LITERATURE REVIEW

Crime prediction has received extensive examination and analysis across various studies, each contributing distinct insights [44], [45], [46]. This section reviews existing literature to offer a detailed overview of the techniques employed in crime detection through sentiment analysis.

A. MACHINE LEARNING

Irawan et al. [47] studied to understand public sentiment concerning enacting the Criminal Code Bill (RKUHP), leveraging sentiment analysis to navigate Twitter (X) data. The objective was to grasp the comprehensive public reaction to

the legalization of RKUHP. Their methodology incorporated the use of N-grams (unigrams, bigrams, trigrams) alongside three algorithms: Naïve Bayes, Classification and Regression Tree (CART), and Support Vector Machine (SVM), with a focus on dissecting the tone of public discourse, while intentionally not delving into the specifics of the resulting data. Oladele et al. [48] delved into the dynamics of social unrest, particularly emphasizing Nigeria's #EndSARS movement, which found its genesis on social media platforms like Twitter (X). Their research methodology involved collecting, preprocessing, and annotating tweets marked with #EndSARS into categories reflecting positive and negative sentiments. A support vector classifier was subsequently utilized to categorize the sentiments encapsulated in these tweets. Saraiva et al. [49] explored the enhancement of crime prediction and prevention by integrating geospatial technologies with data mining and machine learning. By georeferencing official police data, the study pinpointed crime hotspots and spatial patterns via spatial analysis. Machine learning techniques, including lasso regression, random forest, and decision trees, were employed to extract space-time patterns and identify significant crime variables. In addition, tweets related to insecurity were subjected to topic modeling and sentiment analysis. Abbass et al. [50] devised a framework to forecast significant categories of social media offenses, including Cyberstalking, Cyberbullying, Cyber Hacking, Cyber Harassment, and Cyber Scam, leveraging data harvested from social media platforms. This framework is structured around three core modules: the preprocessing of data (tweets), classification model construction, and the prediction process. To develop the prediction model, a combination of Multinomial Naïve Bayes (MNB), K-Nearest Neighbors (KNN), and SVM was employed to segregate the data into distinct crime categories. An N-Gram language model was integrated with these machine learning algorithms to ascertain the most effective 'n' value and evaluate the system's precision at various levels, such as Unigram, Bigram, Trigram, and 4-gram. Vo et al. [51] harnessed Twitter (X) data spanning seven Indian cities from January 2014 to November 2018 to validate the efficiency of their crime detection methodology. They implemented sentiment analysis to delve into user behaviors and utilized a Markov Model-based part-of-speech tagger and Brown clustering for an in-depth tweet analysis. Ristea et al. [52] investigated the spatial interconnections between criminal activities and demographic, socio-economic, and environmental variables with geo-tagged Twitter (X) messages, focusing on game days for basketball and hockey compared to non-game days. This examination involved analyzing the spatial and temporal alignment of crimes and tweets, employing feature selection models to pinpoint essential variables, and adopting localized kernel density estimation for projecting crime occurrences about sporting events and during regular days. Mahor et al. [53] crafted a system specifically pinpointing Cyber Crime Hubs within India, focusing on Jamtara, by analyzing Twitter (X)'s extensive media content repository. This system

delineates criminal alert zones based on reported crime incidents, utilizing a sigmoid kernel and SVM classification to significantly enhance the accuracy of cybercrime trend identification, surpassing the benchmarks set by official government data.

While machine learning methods have significantly advanced issues is the reliance on large volumes of high-quality data; machine learning algorithms require extensive datasets to train effectively, and these datasets must be accurately labeled and free from biases, which is only sometimes feasible in real-world scenarios. Additionally, machine learning models can sometimes act as black boxes, providing little insight into how decisions are made, which can be problematic for accountability and ethical considerations in law enforcement applications. Another challenge is the dynamic nature of crime; as criminal behaviors evolve, models may quickly become outdated, necessitating continuous updates to maintain accuracy. This adaptability issue is compounded by the potential for overfitting, where models may perform well on training data but must generalize to new, unseen data. Furthermore, the diversity and complexity of crime data, including unstructured data such as text from police reports or social media, present significant preprocessing challenges that can affect the performance of machine learning models.

B. DEEP LEARNING

Tam and ÖzgürTanrıöver [4] tried to refine crime prediction methodologies by merging semantic insights from Twitter (X) sentiment analysis with historical crime statistics. They employed a data fusion strategy incorporating a Convolutional Bidirectional LSTM (ConvBiLSTM) model to amalgamate feature vectors from the tweet and crime data into a cohesive representation. This innovative approach was rigorously evaluated using data sets provided by the Chicago Police Department, alongside tweets infused with crime-related terminology from Chicago. It was benchmarked against various crime prediction models, including those based on the BERT framework. Boukabous and Azizi [5] ventured into the realm of crime detection on social media platforms like Twitter (X), employing sentiment analysis enhanced by deep learning (DL) techniques, notably the Bidirectional Encoder Representations from Transformers (BERT), within the field of natural language processing (NLP). Their study adopted a hybrid approach that amalgamated lexicon-based labeling with DL, using BERT. Initially, a lexicon-based method was deployed to label a Twitter (X) dataset with standard and crime-related lexicons, which then facilitated the training of the BERT model, aiming to bolster crime detection through sentiment analysis. Mahajan and Mansotra [6] tackled the evolving strategies of criminals by proposing the analysis of social media data, specifically geo-tagged tweets from five distinct regions in India, to predict crime patterns. Acknowledging the pervasive use of social media, the study sought to exploit the digital footprints left on platforms such as Twitter (X) to glean early warnings

of criminal activities. The research aimed to evaluate crime intensity across various regions through data mining techniques to unearth correlations between tweets and crime patterns and the application of semantic sentiment analysis via Bi-directional Long Short-Term Memory (BiLSTM) and feedforward neural networks. Sandagiri et al. [54] developed a deep learning framework aimed at predicting crimes through the analysis of Twitter (X) data, capitalizing on the tendency of social media users to post information in the field of crime prediction. This comprehensive framework is structured around three pivotal modules: the collection of data, the detection of crime-related content, and the prediction of potential criminal activities, utilizing a Long Short-Term Memory (LSTM) neural network model to forecast crime. This model is specifically designed to tackle the issue of undetected crimes due to the lack of substantial evidence. Sandagiri et al. [55] introduced a crime prediction strategy that leverages data from Twitter (X) and meteorological sources to confront the challenges posed by the scarcity of data and the prevalence of unrecorded crimes. Their methodology is divided into two key modules: the first involves using BERT to pinpoint tweets relevant to crime, and the second employs an Artificial Neural Network (ANN) for crime prediction. Deepak et al. [56] proposed a Bi-LSTM neural network designed to classify various crime types, drawing upon data from Google News and Twitter (X). This process entails a series of steps beginning with data pre-processing, followed by labeling through the application of the Fuzzy c-means algorithm and Term Frequency-Inverse Document Frequency (TF-IDF) vectors, and culminating in feature extraction via GloVe word embeddings. An integral part of their approach is the creation of dynamically generated ontologies with limited human oversight, which significantly enhances the accuracy of crime classification.

Deep learning methods face challenges with imbalanced class distribution and exhibit sensitivity towards hyperparameter configurations. To address these challenges, our study proposes a novel approach that combines the strengths of RL for handling imbalanced classification and ABC algorithms for optimizing hyperparameters. This innovative approach aims to offer a sophisticated, theoretically robust solution that expands the toolkit for crime prediction, making it more versatile and practical in diverse real-world scenarios.

III. BACKGROUND

This section outlines foundational concepts and their brief descriptions.

A. EXPLOITATION IN RL

In RL, navigating the balance between exploration and exploitation is crucial. RL agents refine their actions through environmental interactions, with their behavior influencing these interactions [57]. For enhanced learning, an RL agent must explore to discover rewarding paths. Conversely, the agent must leverage its acquired knowledge for optimal outcomes for reward assurance. Initially, RL agents tend to

explore but gradually focus on exploitation as they gain a better understanding of their environment. Excessive exploration can hinder the development of efficient strategies for exploitation, while too much exploitation might result in less-than-ideal strategies [58].

1) ENTROPY LOSS

As per the literature [59], policy loss in actor-critic RL frameworks is defined by the formula $-A \log(p_i)$, where A signifies the advantage in the actor-critic structure, and p_i denotes the action's probability. Actor-critic methods often incorporate an entropy component into the loss equation to foster exploration. This component reaches its lowest value when every action is equally probable. While the policy aspect aims to optimize reward utilization by enhancing the likelihood of beneficial actions, the entropy component encourages uniform exploration across all actions. The combined policy and entropy loss function in these algorithms is articulated as $-(A - \alpha p_i) \log(p_i) + \alpha \sum_{j,j \neq i} p_j \log(p_j)$, distinguishing the entropy of action i from the overall sum. Here, $\alpha \geq 0$ adjusts the entropy term's magnitude, thereby gradually increasing exploration as α rises. The term $\alpha \sum_{j,j \neq i} p_j \log(p_j)$ acts as a bias, ensuring the probability of the best action never hits one due to the non-minimal entropy term [60].

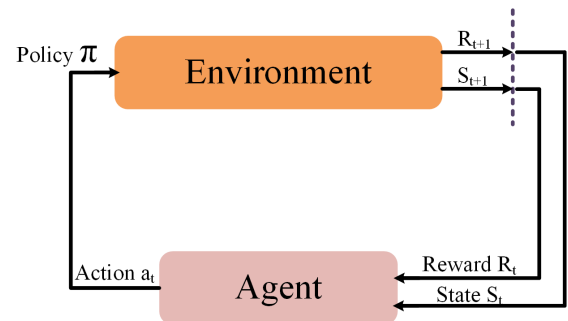


FIGURE 1. Illustration of the MDP framework applied to RL environments.

B. SUPERVISED CLASSIFICATION AS RL

RL environments are commonly structured around the Markov decision process (MDP) framework, which provides a coherent approach to modeling decision-making in environments where outcomes are predictable, as depicted in Figure 1 [61]. Viewing the data classification process through the lens of MDP, each piece of unlabeled data represents a distinct Markov state, and the agent's task of classifying it into the appropriate category is akin to executing a Markov action from a range of possible categories. The feedback for the agent's actions comes in the form of rewards based on the accuracy of the classifications made; higher accuracy yields greater rewards, while errors result in penalties [62]. Utilizing labeled training data, a reward function can be established to aid the agent in anticipating the consequences of its decisions, thereby improving the classification process's efficiency and accuracy [63]. From an

RL standpoint, the task of classification can be viewed either as an episodic MDP, where each data classification initiates a new episode, within which the agent makes a decision and receives immediate feedback in the form of a reward (represented mathematically as states (s_0, s_T) , an action a_0 , and a reward r_0 , or as a continuous MDP, where the classifier navigates through a sequence of state transitions, detailed formally through sequences of states $(s_0, s_1, \dots)$, actions $(a_0, a_1, \dots)$, and rewards $(r_0, r_1, \dots)$. When setting α to 1 and identifying i as the index for the correct class, the policy and entropy loss within a classification context is expressed as $-\log(p_i) + \alpha \sum_j p_j \log(p_j)$. If α is set to 0, removing the entropy component, the policy loss simplifies to the cross-entropy loss. Introducing the advantage term, A , and a hyperparameter γ to modify p_i , about A , the focal loss is formulated as $-(A - \alpha p_i)^\gamma \log(p_i)$, integrating the concept into actor-critic RL algorithms [64].

C. ARTIFICIAL BEE COLONY

The ABC algorithm [41] is built on two primary components: food sources and artificial bees. The algorithm simulates the foraging behavior of bees, where the locations of food sources represent potential solutions to the optimization challenge. Within the ABC framework are three bee roles: employed bees, onlooker bees, and scout bees. Employed bees are tasked with exploring the vicinity of known food sources for richer nectar sources and relaying this information to the onlooker bees within a designated dance area. The colony is structured so that the number of employed bees matches the number of onlooker bees, comprising half of the total colony size. Each employed bee is associated with one hive, equating the number of hives to the number of employed bees. Following cues from employed bees, onlooker bees also probe their immediate surroundings for superior food sources. Employed bees unable to enhance their food sources after a predetermined number of attempts transition into scout bees, embarking on entirely new searches [65].

The ABC algorithm unfolds over four iterative phases: initialization, employed bee phase, onlooker bee phase, and scout bee phase, continuing until a stopping condition is met. The initialization phase sets up food sources based on bee positions within the search domain, as per Equation 1, where x_{min}^j and x_{max}^j denote the minimum and maximum bounds for solution j , respectively.

$$x_i^j = x_{min}^j + rand(0,1)(x_{max}^j - x_{min}^j) \quad (1)$$

During the employed bee phase, new potential food sources are identified and evaluated to replace lesser-quality sources, as described in Equation 2 [66].

$$v_i^j = x_i^j + \varphi_i^j(x_i^j - x_k^j) \quad (2)$$

Here, k is a randomly chosen index within the colony's size, φ_i^j is a random number between -1 and 1 , and v_i represents the newly discovered food source derived from modifying v_i .

The onlooker bee phase involves employed bees sharing their findings with onlooker bees, who then probabilistically

Algorithm 1 The ABC Algorithm

Input:

D : Dimensionality of the solution space

BN : Size of the bee population

limit: Cycle threshold for scout bee conversion

$MaxItr$: Largest amount of algorithm cycles

Create the solution set $X = [x_1, x_2, \dots, x_{BN}]$ via Equation (1)

Set iteration count: $Itr = 1$

While $Itr \leq MaxItr$ Do

/* Employed Bee Phase */

For $i = 1$ To BN Do

Generate new candidate solution x_{new} using Equation (2)

Evaluate fitness f_{new} of x_{new}

If f_{new} is better than f_i Then

Replace x_i with x_{new}

End If

End For

/* Calculation of Probabilities for Onlooker Bees */

Compute probability p_i for each solution x_i in X using Equation (3)

/* Onlooker Bee Phase */

For $i = 1$ To BN Do

If $random(0, 1) < p_i$ Then

Generate new candidate solution x_{new} using Equation (2)

Evaluate fitness f_{new} of x_{new}

If f_{new} is better than f_i Then

Replace x_i with x_{new}

End If

End If

End For

/* Scout Bee Phase */

Identify any abandoned solution

If an abandoned solution exists Then

Replace with a new solution using Equation (2)

End If

/* Track Best Solution */

Update x_{best} if any x_i in the current population is better

/* Update Iteration Counter */

Increment Itr by 1

End While

Output: x_{best}

choose and assess solutions based on quality, potentially updating the solution pool. The selection probability for a new solution by an onlooker bee is given by Equation 3, where $fit(x_i)$ signifies the fitness of the i^{th} solution [21].

$$p_i = \frac{fit(x_i)}{\sum_{n=1}^{BN} fit(x_n)} \quad (3)$$

Scout bees come into play in the final phase to avert local optimization pitfalls. Solutions that do not show improvement

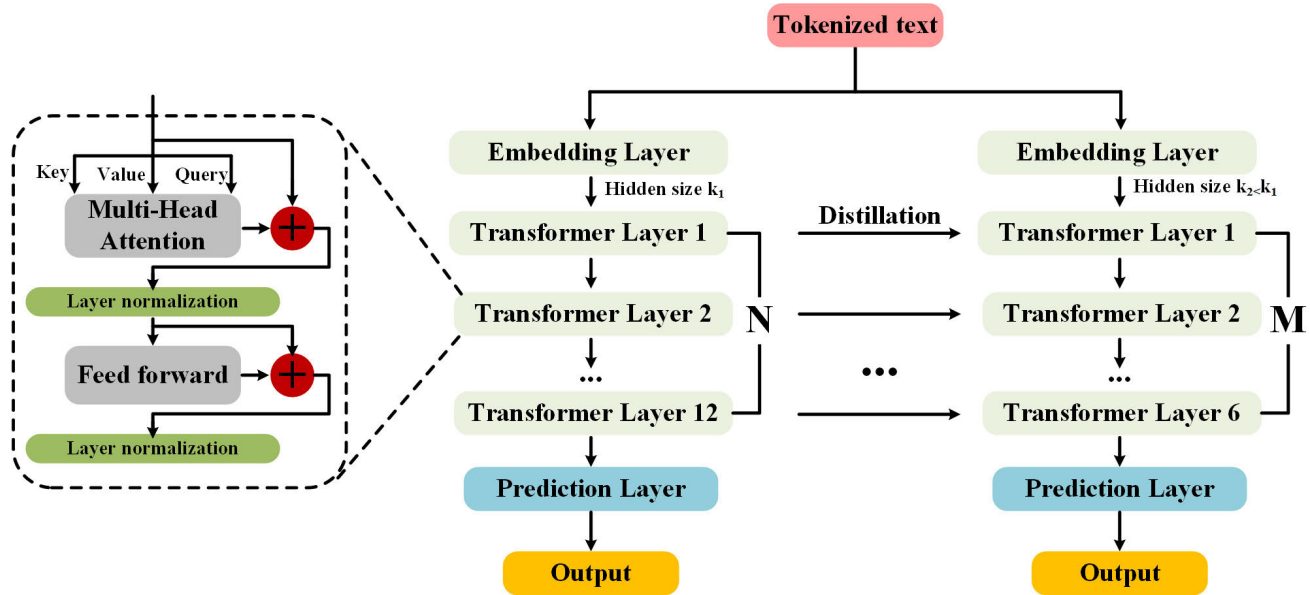


FIGURE 2. The DistilBERT model architecture and components.

over several cycles are abandoned in favor of new searches, effectively resetting the food source as per Equation 2. A comprehensive presentation of the ABC algorithm is explained in Algorithm 1.

D. DistilBERT

DistilBERT [67] is a streamlined version of the original BERT model, achieved through a technique known as knowledge distillation. This process involves training a smaller, more efficient model—the student—to mimic the behavior of a larger, pre-trained model known as the teacher. The training of the student model involves minimizing a distillation loss, which is based on matching the softened output probabilities of the teacher model, as illustrated in Equation 4 [67].

$$L_{ce} = \sum_i t_i * \log(s_i) \quad (4)$$

Here, t_i and s_i represent the output probabilities from the teacher and student models, respectively. Despite its reduced size, DistilBERT maintains much of BERT's original efficacy, being 40% smaller and 60% faster yet retaining 97% of BERT's language understanding capabilities. This reduction is primarily achieved by halving the number of layers in the model from 12 to 6, resulting in DistilBERT having 66 million parameters compared to BERT's 110 million.

Training DistilBERT was notably more resource-efficient, requiring only 3.5 GPU days on eight V100 GPUs, compared to the 12 GPU days needed for BERT under the same conditions. Both models were trained on a 16 GB dataset compiled from English Wikipedia and the Toronto book corpus. DistilBERT's training efficiency is further enhanced by using large batch sizes combined with gradient accumulation, allowing for effective parameter updates by aggregating gradients from several mini-batches. Additionally, DistilBERT's

training process is streamlined by omitting the next-sentence prediction and segment embedding tasks that were part of BERT's training regime. Instead, it adopts a dynamic masking approach during inference, which replaces BERT's static masking method. This adaptation allows for improved model performance in practical applications [67]. Figure 2 shows the DistilBERT model architecture and components.

IV. PROPOSED SCHEME

Figure 3 presents the architecture of our proposed model for semantic analysis. The model takes as input a sentence $D = [w_1, w_2, \dots, w_n]$, where each w_i represents the individual words in the sentence, and n is the sentence's maximum word length. This input sentence is then processed through the BERT model. The output from the DistilBERT model is a set of word embeddings, represented as a matrix $M = [e_1; e_2; \dots; e_n]$, where each e_i corresponds to the embedded vector of the word w_i . To extract sentence features, the embedding matrix M is passed through three dilated convolution layers running in parallel. Each layer independently extracts a set of features from the sentence, providing a comprehensive analysis from different perspectives. After the convolution process, max pooling is applied to each feature set, highlighting the most prominent features while reducing the subsequent layers' computational demands. The features selected by the max pooling process are then fed into the Multi-Layer Perceptron (MLP) network, which is responsible for the final classification task. The MLP network categorizes each sentence into one of two classes: crime or non-crime. However, given the predominance of non-crime classifications in typical datasets, the model faces challenges with imbalanced classification, which can adversely affect overall performance. To mitigate this imbalance, we integrate

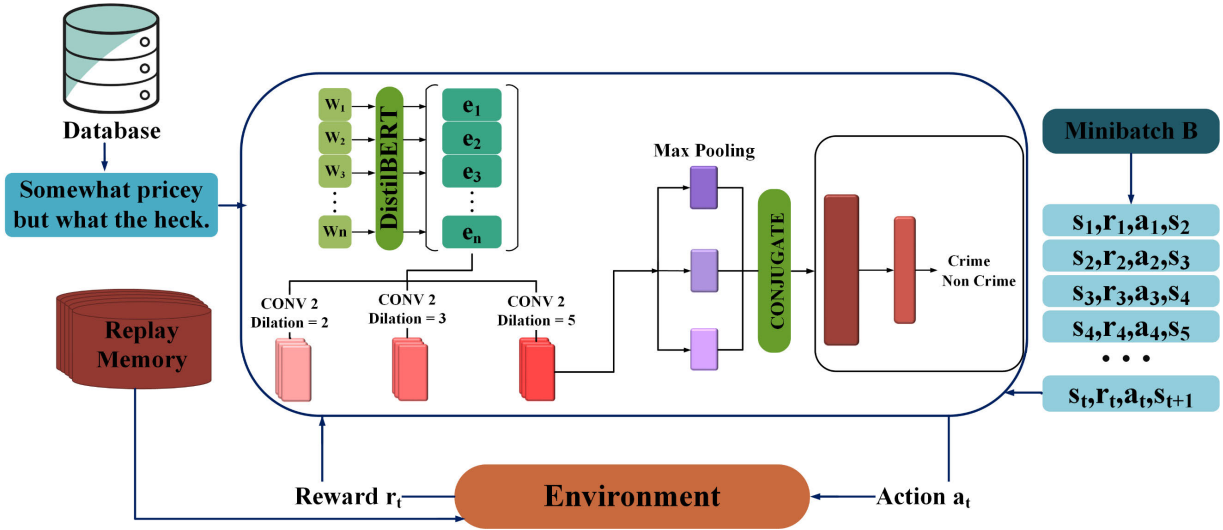


FIGURE 3. The proposed model architecture is composed of a series of interconnected steps. Initially, at the outset of the process, a sample instance (s_t) is selected from the dataset, which is then introduced into the realm of the RL environment. This selected sample is promptly input into the neural network. Subsequently, the prediction made by the network (action a_t) is relayed back to the RL environment. This action results in the retrieval of the next sequential image (s_{t+1}) and its corresponding reward (r_t). These components, encapsulated as $\{s_t, a_t, r_t, s_{t+1}\}$, are meticulously logged within the replay memory. As the replay memory becomes populated with an array of such data sets, these sequences are randomly selected, which then informs the tuning of the network's parameters. This loop is executed repeatedly, enabling the network to refine its classification accuracy of the input images progressively. The process is deemed complete when the predetermined count of episodes is fulfilled, signaling the end of the algorithmic sequence.

an RL-based algorithm and scope loss function within the model. This algorithm functions as a sequential decision-maker, specifically designed to navigate the complexities of imbalanced classification, enhancing the model's ability to effectively distinguish between crime-related and non-crime-related sentences.

The decision to employ an RL algorithm within our semantic analysis model is predicated on the unique capabilities of RL to handle the challenges of imbalanced classification. Traditional classification models often falter when faced with datasets where one class significantly outnumbers another, leading to a bias towards the majority class and poor generalization to minority class instances.

This imbalance is particularly pronounced in semantic analysis tasks where non-crime-related sentences dominate, potentially skewing the model's predictions. RL, with its dynamic decision-making framework, offers a solution to this issue. In RL, the model, or "agent," learns to make a series of decisions (actions) based on the state of the environment (input data) and the feedback received (rewards). By crafting a reward system that preferentially values the accurate classification of the minority class, our RL-based model incentivizes the identification of less common, crime-related sentences. This approach ensures that the model does not neglect the minority class, thus overcoming the skewed performance that plagues conventional models in imbalanced scenarios. Furthermore, the RL framework allows for continuous learning and adaptation, which is critical in the ever-evolving online content landscape where new terminology and expressions emerge regularly. The RL agent

iteratively refines its policy—the strategy of selecting actions based on states—to achieve a more balanced classification performance. This adaptability is crucial for maintaining high accuracy and robustness, regardless of the varying distributions of classes within the training data.

Scope loss is chosen for its proficiency in managing the exploration-exploitation dilemma within RL frameworks, as it fine-tunes gradients to avert performance deterioration due to excessive exploitation and data disparities. Imposing a ceiling on the rate at which an agent shifts towards exploitative behavior subtly fosters exploration, particularly when the agent's action certainty is elevated. This quality renders scope loss adept at encouraging exploration while safeguarding against performance degradation. Moreover, its implementation is straightforward as it does not necessitate hyperparameter tuning and has demonstrated superior results to alternative loss functions in various RL contexts.

A. SCOPE LOSS

During the model's training phase, we employ a novel loss function known as scope loss, inspired by FL and policy and entropy losses. Similar to FL, scope loss addresses class imbalance by adjusting the gradients for each class based on their prediction confidence [68]. Additionally, like policy and entropy losses, scope loss encourages exploration by penalizing certain policies. As shown in Equation 5, scope loss for the actor-critic framework uses $A - \alpha p_i$ as a scaling factor, where α is a hyperparameter that modulates p_i in proportion to the size of A . This adjustment ensures a balanced learning approach, promoting accuracy and exploratory behavior

during model training.

$$\text{Scope Loss(Actor Critic)} = -(A - \alpha p_i) \log(p_i) \quad (5)$$

Unlike policy and entropy loss, which broadly encourage exploration by randomly incorporating exploratory actions into the agent's behavior, scope loss promotes exploration more strategically by controlling how quickly an agent can transition to exploitative actions [69]. When the advantage term A is positive, a higher certainty in action probability p_i results in smaller gradients, effectively delaying premature exploitation. Notably, scope loss can be simplified by omitting the bias term in its equation, theoretically enabling the agent to reach an optimal policy more efficiently. Additionally, scope loss can be redefined by setting the parameter $\gamma = 1$ in its base equation, eliminating the need for additional hyperparameters. For supervised classification tasks, the complexity is further reduced by setting both A and α to 1, as illustrated in Equation 6.

$$\text{Scope Loss(Supervised)} = -(1 - p_i) \log(p_i) \quad (6)$$

Determining the most effective value of α in reinforcement learning poses a significant challenge, as the distribution of the advantage term A fluctuates across different environments and stages of the training process. The value of α must be small enough to ensure that the advantage term predominantly influences the scope term. Rather than manually adjusting α for each unique environment, a fixed α can be used in conjunction with z-score batch normalization of the advantages, as described in Equation 7. This approach helps establish a universally effective α value, making it applicable across various environments and throughout the training process [70].

$$A = \frac{A - \mu_A}{\sigma_A} \quad (7)$$

1) SCOPE LOSS FOR PROXIMAL POLICY OPTIMIZATION

We have integrated scope loss into actor-critic RL algorithms and supervised classification models. Recognizing the effectiveness of proximal policy optimization (PPO) algorithms in distributed settings, we are now exploring how to apply scope loss within PPO frameworks. To begin, we consider the standard policy loss function typically used in PPO:

$$\begin{aligned} \text{Policy Loss (PPO)} \\ &= -\min\left(\frac{p_i}{p_i^k} A^k, g(\epsilon, A^k)\right), \text{ where } g(\epsilon, A) \\ &= \begin{cases} A(1 + \epsilon) & \text{if } A \geq 0 \\ A(1 - \epsilon) & \text{if } A < 0 \end{cases} \end{aligned} \quad (8)$$

Here, the index i represents the chosen action, while k identifies the worker model that updates the primary model using its local experiences. The hyperparameter ϵ in PPO, known as clipping sensitivity, regulates the permissible deviation between the policies of the worker and the primary model before the loss gradient is clipped. To enhance the traditional

policy and entropy loss function employed in PPO, we also incorporate an entropy term:

Policy and Entropy Loss (PPO)

$$\begin{aligned} &= -\min\left(\frac{p_i}{p_i^k} A^k, g(\epsilon, A^k)\right) \\ &+ \alpha \sum_j p_j \log(p_j) = -\min\left(\frac{p_i}{p_i^k} A^k, g(\epsilon, A^k)\right) \\ &+ \alpha \times p_i \times \log(p_i) + \alpha \sum_{j,j \neq i} p_j \log(p_j) \end{aligned} \quad (9)$$

Including the min term prevents straightforward factorization from deriving the scope factor. By setting $\alpha = 0$, we adapt scope loss specifically for use within PPO algorithms.

$$\begin{aligned} \text{Focal loss (PPO)} &= -\min\left(\frac{p_i}{p_i^k} A^k, g(\epsilon, A^k)\right) \\ &+ \alpha p_i \log(p_i) \end{aligned} \quad (10)$$

When the expression $\frac{p_i}{p_i^k} A^k$ yields a minimum, we adjust the equation to highlight the scope scaling factor particularly relevant to PPO [71].

$$-\frac{p_i}{p_i^k} A^k + \alpha p_i \log(p_i) = -\frac{p_i}{p_i^k} (A - \alpha p_i^k \log(p_i)) \quad (11)$$

In the realm of PPO applications, the scope scaling factor, represented as $(A - \alpha p_i^k \log(p_i))$, significantly influences the gradient adjustments by considering the cross-entropy between the policies of the main model and the worker model. This factor modifies the gradients in various ways:

- When the primary model shows high confidence (p_i), and A is positive, the gradient is reduced to prevent premature exploitation.
- If the worker model's confidence (p_i^k) is lower than that of the main model (p_i) and A is positive, the gradient is moderated to minimize divergence between the models.
- Conversely, if the worker model's confidence (p_i^k) surpasses that of the primary model and A is negative, the gradient is similarly reduced to decrease divergence between the models.

Finally, we adapt the FL concept to integrate seamlessly with the PPO framework as follows:

$$\begin{aligned} \text{Focal loss (PPO)} &= -\min\left(\frac{p_i}{p_i^k} A^k, g(\epsilon, A^k)\right) A \\ &+ 2\alpha \min\left(\frac{p_i}{p_i^k} A^k, g(\epsilon, A^k)\right) p_i \\ &- \alpha^2 p_i^2 \log(p_i) \end{aligned} \quad (12)$$

2) OVERALL ALGORITHM

In our model, we define the state, action, and reward as follows:

- State s_t : This represents the sample observed at step t , serving as the basis for the network's decisions.

Algorithm 2 Pseudo-Code of Training the Model**Input:** $D = \{(x_1, y_1), (x_2, y_2), \dots, (x_T, y_T)\}$: training data E : number of iterations α : learning rateInitialize replay memory B **For** each episode $e = 1$ to E **Do**Randomize the order of the dataset D Set the initial state $s_1 = x_1$ **For** each iteration $t = 1$ to T **Do**Select action a_t according to policy derived from Equation 13Calculate reward r_t for the current transition using Equation 14Determine the next state s_{t+1} from D Store the transition (s_t, a_t, r_t, s_{t+1}) in replay buffer B **End For****For** each optimization iteration $i=1$ to N **Do**Draw a random minibatch of transitions (s_k, a_k, r_k, s_{k+1}) from B Update the policy π by maximizing the off-policy PPO clip objective from Equation 12:

$$\theta \leftarrow \theta_{old} + \alpha \nabla_{\theta} \text{Focal loss (PPO)}$$

End For**End For**

- Action a_t : We adopt a more sophisticated exploration method from the traditional ϵ -greedy strategy often employed in Q-learning, which typically favors immediate rewards but may neglect ample exploration in environments with sparse data. Utilizing the upper confidence bound approach, this method combines cumulative knowledge with a reduced reliance on random choice. The following equation determines the action:

$$a_t = \arg \max_a [Q_t(a) + \phi \sqrt{\frac{\ln n'}{n'_a}}] \quad (13)$$

Here, n' indicates the total number of times all actions have been selected, n is the count of times action a has been chosen, and ϕ is an exploration constant.

- Reward r_t : Rewards are given for each classification to guide the network towards accurate categorization. The reward is calculated as follows:

$$r_t(s_t, a_t, y_t) = \begin{cases} +1, & a_t = y_t \text{ and } s_t \in D_O \\ -1, & a_t \neq y_t \text{ and } s_t \in D_O \\ \lambda, & a_t = y_t \text{ and } s_t \in D_N \\ -\lambda, & a_t \neq y_t \text{ and } s_t \in D_N \end{cases} \quad (14)$$

where D_O and D_N represent the minority and majority classes, respectively. Correct or incorrect classifications in the majority class result in rewards of $+\lambda$ or $-\lambda$. This system prioritizes the precise classification of minority class samples by offering greater rewards for such instances. By incorporating the majority class and a tunable reward parameter λ ($0 < \lambda < 1$), we introduce complexity into the reward structure, which fine-tunes the network's attention between frequent and infrequent classes.

As outlined in Algorithm 2, our method employs an extensive training protocol that enables the agent to engage in continuous learning and adaptation until it converges into an optimal policy. At the heart of our model is a self-enhancing decision-making process designed to ensure that decisions are optimized for the agent's benefit. The impact of each agent action is quantitatively appraised using Equation 14, which facilitates the evaluation of the success of our implemented strategies. Our methodology entails a rigorous iterative process, conducted over E iterations—set at 256 for this investigation. Following each iteration, we document the policy network's parameters, which shed light on the learning trajectory and chart its development. This meticulous iterative method enables a comprehensive performance assessment of the model. Through such in-depth analysis, we can make well-informed decisions for further enhancements and adjustments to the model, fostering continuous improvement and fine-tuning of our strategic approach.

B. OPTIMIZATION OF HYPERPARAMETERS

Optimizing hyperparameters is a critical process in deep learning, acting as a key driver that can significantly improve the performance of neural networks. Fine-tuning hyperparameters, which include learning rate, batch size, the number of network layers, and the choice of activation functions, can markedly increase a model's predictive capabilities and efficiency during training [72]. Properly optimized hyperparameters address common issues such as overfitting and underfitting, enhancing the model's ability to generalize to new data. Additionally, in the context of deep learning's intensive computational requirements, hyperparameter

optimization contributes to more effective use of resources, saving both time and economic investments.

TABLE 1. Space of parameters for optimizing hyperparameters.

Hyperparameter	Values	Best
Learning rate	[0-1]	0.06
Batch size	16 to 256	30
Epoch	64 to 512	258
Activation function	ReLU, Leaky ReLU, Linear, Tanh, Sigmoid	ReLU
Dropout rate	[0-1]	0.5
Number of layers in MLP	1,2,3,4,5,6	3
Number of CNNs	1,2,3,4,5,6	4
γ	[0-1]	0.4

Table 1 delineates the hyperparameters tailored for optimization in this study. Each hyperparameter was assigned a range of plausible values, informed by the scopes documented in the current literature on deep learning for sentiment analysis for crime prediction. These predefined ranges were then applied as limits within the execution of the ABC algorithm.

For hyperparameter optimization in this study, we adopt the random critical approach, a technique devised initially for genetic algorithms [73]. This approach involves encoding a population of T D-dimensional numerical vectors labeled as $p_1, p_2, \dots, p_T$. Each vector within this population encapsulates a potential solution, correlating to a set of model hyperparameters via a mapping function known as the random key. Considering C hyperparameters to optimize (where C equals 8 for the proposed model as per Table 1), each hyperparameter c (ranging from 1 to C) is represented by D_c dimensions, with D_c equalling 1 for continuous hyperparameters. Thus, the total dimension D is the sum of all D_c . Each vector p_i is divided into C segments, with D_c dimensions apiece, where each dimension in D signifies a specific hyperparameter value. For categorical hyperparameters, a mapping from the numerical vector's D_c -dimensional block to a corresponding vector MAP_c occurs in the following manner: the D_c values are ordered, and the position of the first element determines the index for MAP_c , from which the relevant classification value is chosen. An illustration of the random critical method in action is depicted in Figure 4.

1) THE PROPOSED ABC

In the standard ABC algorithm, artificial bees independently select and modify a food source position to generate a new candidate position. If the new solution's fitness surpasses that of the current one, the current solution is replaced. Otherwise, the original solution remains unchanged. Specifically, in an optimization problem with D dimensions, a single dimension is selected randomly for modification, and in each iteration, the solution with the better outcome prevails.

As dictated by Equation 2, the newly produced solution v_i^j relies only on two factors: x_i^j and x_k^j . This dependency

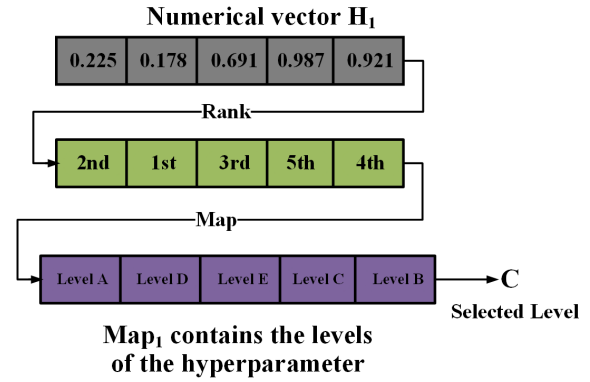


FIGURE 4. The random key method employed for converting numerical vectors into distinct levels of hyperparameters.

renders the food source v_i^j quite volatile, alternating between values higher or lower than the existing food source. Within the ABC algorithm's framework, finding food sources that demonstrate superior fitness values is crucial. To generate consistently improved food sources, the algorithm uses the fitness data obtained from reciprocal interactions between the current and proximate food sources:

$$v_i^j = \begin{cases} x_i^j + \phi_i^j (x_k^j - x_i^j), & \text{Fit}_i < \text{Fit}_k \\ x_k^j + \phi_i^j (x_i^j - x_k^j), & \text{Fit}_i \geq \text{Fit}_k \end{cases} \quad (15)$$

Here, Fit_i and Fit_k denote the fitness values of the current and adjacent food sources, respectively, while ϕ_i^j is a uniformly distributed random number within the range $[0, F]$, where F is a non-negative constant known as the mutual learning factor. The value v_i^j is influenced by both the spatial position and the fitness value of the food sources. The new solutions' fitness values trend towards the superior source by comparing the current and neighboring sources. If the current food source is of better fitness, the candidate solution will move towards an enhanced position; if not, it leans towards the adjacent source. This reciprocal learning mechanism ensures the production of superior candidate solutions. The role of parameter F is to balance the adjustments among food source positions. Maintaining F as a non-negative value is essential for steering toward enhanced solutions. As F increases from zero to a specific threshold, the perturbations affecting the food source positions reduce, meaning the fitness of the new solution becomes more akin to the higher fitness source. However, a tremendous value of F can dampen the algorithm's ability to exploit and explore, thus impacting the overall performance.

V. EMPIRICAL EVALUATION

In this section, we describe the dataset, detailing its features and scope as used in our study. This is followed by an explanation of the metrics, where we specify the criteria and measurements employed to assess our models' performance. Subsequently, we introduce the pre-processing

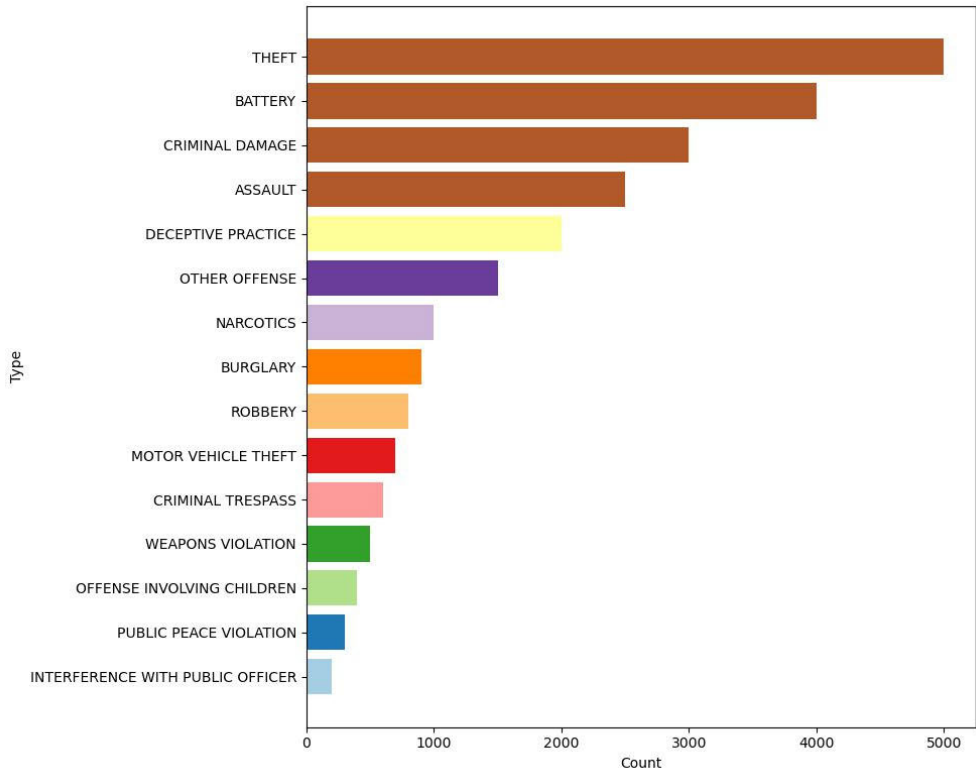


FIGURE 5. Distribution of crime types in Chicago.

methods applied to the data. The section concludes with the presentation of results, highlighting the key findings from our analysis and model evaluations and discussing their significance in the context of our research objectives.

A. DATASET

This study used distinct datasets to evaluate the proposed model: The Chicago Police Department data and Chicago-related tweets. We chose these datasets to comprehensively analyze crime from official records and public perception, enabling a dual-perspective approach to crime prediction. Moreover, we chose Chicago because of its high crime rates and the availability of detailed crime data and social media activity, making it an ideal urban laboratory for studying crime patterns and public reactions [4].

The first dataset includes data on five types of crime incidents: theft, battery, burglary, robbery, and motor vehicle theft. This data, taken from the city of Chicago, spans from September 2019 to July 2024 and was accessed through the Chicago data web portal [74]. It contains 22,418 reported incidents of crime. As shown in Figure 5, an analysis of the dataset shows a significant variation in the frequency of different types of crimes. Theft is the most common, with over 6,000 incidents, making up 26.13% of the total, followed by battery at 20.15%.

The second dataset consists of tweets related to crime-related terms. It was gathered from Twitter (X) during the

same period, using the GetOldTweets Python library, and includes 512,361 tweets. These tweets range from public discussions about crime to general tweets from the Chicago area. Twitter (X) data helps capture the public’s immediate reactions and discussions surrounding these crimes, providing contextual insights not available in traditional crime reports [9]. Given the high incidence of theft and other related crimes in the crime data, these categories, such as battery, burglary, robbery, and motor vehicle theft, were explicitly targeted in our crime prediction model. We thoroughly analyzed tweets containing terms and synonyms related to these crimes. Words like ‘theft,’ ‘crime,’ ‘battery,’ ‘steal,’ ‘burglary,’ ‘robbery,’ and ‘motor vehicle theft’ were used to filter relevant tweets from the larger dataset for our study. This process led to identifying and extracting 20,562 tweets for further analysis. Examples of the types of tweets collected for this study are presented in Table 2.

B. METRICS

Our issue revolves around imbalanced classification and choosing critical evaluation metrics. We employ measures such as Accuracy, Precision, Recall, and F1 Score, defined as follows [25]:

- Accuracy: This metric calculates the ratio of correctly predicted instances to the total number of predictions. Although helpful, its reliability may be compromised in

TABLE 2. Sample tweets related to crime from the Chicago area.

Date	Time	Tweets
2019-09-12	23:34:21	Employee theft, particularly at distribution canters, is a constant threat to supply chains. Wemobilize teams in escort vehicles for uninterrupted travel and constant surveillance. Our technology & expertise in freight security ensures that your supply chain is always protected
2019-09-12	13:25:35	Blue Star Security CEO Michael Sapracone discusses the alarming and increasing trend of employee theft in his latest article
2019-09-24	15:54:34	Why aren't victims of theft also "workers"?
2019-09-23	14:34:10	Creighton got an absolute steal
2019-09-29	03:11:13	Haha. That's a good point. I'm gonna steal that

imbalanced datasets, where a model might accurately predict the majority class but perform poorly on the minority class [75].

- **Precision:** This is the proportion of true positive predictions about the total number of positive predictions made. It becomes essential in situations where the consequences of false positives are significant. Within imbalanced datasets, high precision for the minority class suggests that the model efficiently identifies genuine instances of the less common class.
- **Recall (Sensitivity):** This metric determines the proportion of true positive predictions out of all positive instances. It is vital for imbalanced datasets as it indicates the model's capability to recognize the minority class.
- **F1 Score:** The F1 Score is the weighted average of Precision and Recall, providing a balance between the two. In the context of imbalanced datasets, the F1 Score is a more appropriate measure of model performance, addressing the limitations of the Accuracy metric by accounting for both false positives and false negatives [21].

These metrics are mathematically represented as [19]:

$$Accuracy = \frac{TP + TN}{Total\ number\ of\ samples} \quad (16)$$

$$Precision = \frac{TP}{TP + FP} \quad (17)$$

$$Recall = \frac{TP}{TP + FN} \quad (18)$$

$$F - measure = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (19)$$

Here, TP (True Positives) are instances correctly identified as positive, TN (True Negatives) are instances correctly identified as negative, FP (False Positives) are negative instances incorrectly labeled as positive, and FN (False Negatives) are positive instances incorrectly labeled as negative.

C. PRE-PROCESSING

Pre-processing involves filtering unnecessary and irrelevant words, which lightens the computational load and boosts processing speed [76]. Moreover, pre-processing helps mitigate overfitting by removing noise and extraneous data that could lead the model to pick up patterns specific to the training set. By concentrating on more significant features, pre-processing enhances the model's ability to generalize to new, unseen data, thereby reducing the risk of overfitting [77]. In this article, two techniques were employed for pre-processing:

- **Stop-word removal:** Words like 'or' and 'with' often lack semantic value due to their frequent occurrence across documents. Removing these words is vital for improving the accuracy of crime detection.

Stemming: Stemming reduces words to their base form, such as converting 'looking' to 'look.' This process ensures that word variations are treated consistently, which aids in better feature extraction and model performance.

D. MAIN RESULTS

We employed a 5-fold stratified cross-validation method to evaluate our models' effectiveness and consistency. Stratified cross-validation helps reduce overfitting by ensuring that each fold of the dataset maintains the same class distribution as the entire dataset [78]. This approach allows the model to train and validate on diverse subsets, promoting better generalization and minimizing the risk of learning patterns specific to only certain data portions. This approach divides the dataset into five equally sized partitions that reflect the original distribution of the data. In each round of validation, one partition is used for testing while the remaining four serve as the training set. This process is repeated five times, with each partition used once as the test set, ensuring that all data is used for training and testing. This systematic procedure ensures thorough data utilization for training and evaluation of the model. The performance metrics are averaged over the five iterations to provide a reliable measure of the model's performance. The results are reported as $M \pm S$, where M represents the mean performance across all iterations, and S indicates the standard deviation, illustrating the variability of the results.

The proposed crime prediction model was compared against ten deep learning methods: LSTM [54], GSCA [3], TwitterGuard [5], ConvBiLSTM [4], SocioCrimAnalytix [9], CNN-LSTM [79], DAC-BiNet [80], LSA [81], LSTMTwitter [82], and MDL [4]. Additionally, we juxtaposed the model against three derivative models: 'Proposed w/o RL,' 'Proposed w/o SLF,' and 'Proposed w/o ABC,' which does not use RL, scope loss, or ABC for training the model. Tables 3 and 4 summarize these comparative results for two datasets.

When analyzing the Chicago crime dataset, the proposed model exhibited the highest improvement across all performance metrics: accuracy of 96.411%, precision of 93.485%, recall of 93.188%, and F-measure of 93.944%. This marks a

significant enhancement over the top-performing traditional model, MDL, which posted figures of 93.674% for accuracy, 90.729% for precision, 90.646% for recall, and 90.125% for F-measure. The increments in accuracy (2.737%), precision (2.756%), recall (2.542%), and F-measure (3.819%) suggest that the integration of RL, SLF, and ABC significantly boosts the proposed model's efficiency. These components improve the model's ability to handle imbalanced data and refine learning processes more adeptly than other models. Examining other high-performing models, such as LSTMTwitter and DAC-BiNet, which recorded accuracies of 92.315% and 91.532%, respectively, the proposed model surpasses the with improvement rates of about 4.096% and 4.879% in accuracy, respectively. These models leverage advanced features like LSTM's time-series analysis and DAC-BiNet's attention mechanisms, which are effective but still not as comprehensive as the proposed model's approach. This suggests that while these features are beneficial, the additional enhancements in the proposed model, such as RL for better generalization and ABC for optimal hyperparameter tuning, provide a critical advantage. In contrast, simpler models such as LSTM and GSCA, which achieve accuracies of 79.495% and 80.687%, respectively, show much lower performance. The proposed model demonstrates an accuracy improvement of approximately 16.916% over LSTM and 15.724% over GSCA, underscoring the limitations of models that do not incorporate newer or hybrid methodologies for managing complex, real-world datasets. Further comparison of the proposed model with its derivatives—such as 'Proposed w/o RL,' 'Proposed w/o SLF,' and 'Proposed w/o ABC', highlights substantial enhancements. For example, including RL results in a 14.1% increase in accuracy and a 19.2% rise in F-measure compared to the version without RL, emphasizing RL's role in effectively managing imbalanced data and enhancing the model's robustness to ensure that less frequent crime types are not overlooked. Similarly, the absence of SLF leads to an 11.3% drop in precision, underscoring SLF's critical role in balancing known data with exploring new data. The absence of ABC results in an 8.3% decline in accuracy, demonstrating how effectively the ABC algorithm tunes hyperparameters to optimize performance.

For the Chicago tweets data, the proposed model again surpasses all compared models, with a notable 2.5% increase in accuracy and a 4.6% improvement in the F-measure compared to MDL, the next best-performing model. The proposed model's effectiveness in handling imbalanced classification is particularly evident here, as social media data often presents significant class imbalances, with some crime-related topics being much less represented than others. The model's ability to integrate multiple data sources and dynamically adjust to varying data characteristics proves highly effective in managing this type of data, which is often noisy and unstructured. The model's superior performance over other advanced models like LSTMTwitter and LSA can be attributed to its advanced architecture, which effectively balances the trade-offs between feature extraction and model generalization,

particularly in imbalanced classes. The improvements are equally significant when comparing the proposed model with its derivative versions on the tweets dataset. The full model's accuracy exceeds that of 'Proposed w/o RL' by 15.4%, underscoring the importance of RL in adapting to the complex, imbalanced nature of social media data. The 'Proposed w/o SLF' version shows a 13.1% lower F-measure, indicating that SLF is crucial for ensuring the model efficiently balances the exploitation of known data with the exploration of new, potentially valuable data, which is vital for addressing dataset imbalances. The 'Proposed w/o ABC' model also falls short, with an 11.4% decrease in accuracy, reinforcing the value of ABC in optimizing hyperparameters for enhanced model performance.

To assess the statistical significance of our proposed model's superiority over existing models, we conducted paired t-tests on the results. We calculated p-values for each comparison across four key metrics: Accuracy, Precision, Recall, and F-measure. The analysis of these p-values reveals that our proposed model consistently achieves statistically significant improvements over the existing models across all metrics. For the Chicago crime dataset, p-values for the accuracy, precision, recall, and F-measure differences between our model and the next best-performing model, MDL, are all below 0.01, indicating strong statistical significance. Confidence intervals for these metrics further reinforce our findings, with the proposed model consistently demonstrating superior performance. For instance, the 2.7% improvement in accuracy and the 3.8% increase in the F-measure over MDL are statistically significant, underscoring the robustness of the proposed model's architecture in managing complex crime data. Similarly, in the analysis of the Chicago tweets dataset, p-values comparing our model to top existing models like MDL and LSTMTwitter also fall below 0.01, confirming the statistical significance of our results. The confidence intervals show that these improvements are consistent and not due to random variation. The statistical analysis reveals even more pronounced improvements when comparing the proposed model to its derivative versions, 'Proposed w/o RL,' 'Proposed w/o SLF,' and 'Proposed w/o ABC'. The p-values for accuracy, precision, recall, and F-measure differences between the full proposed model and its derivatives are well below 0.001. This underscores the critical importance of RL for handling imbalanced data, SLF for optimizing the balance between known and new data, and ABC for effective hyperparameter tuning. Overall, the statistical significance of these results, evidenced by low p-values and narrow confidence intervals, strongly supports our proposed model's superiority over existing models and their ablated versions. This confirms that our model's enhancements are effective and statistically robust across various datasets and scenarios.

Figure 6 depicts the training and validation loss curves over 300 epochs for the Chicago crime data and the Chicago Twitter (X) data. For the Chicago crime data, the loss trajectories display a consistent and synchronized decline in training and validation losses, with the curves closely aligned throughout

TABLE 3. Comparative performance metrics of the proposed model relative to advanced models and ablated studies on the Chicago crime data.

Model	Accuracy	Precision	Recall	F-Measure
LSTM [54]	79.495 ± 0.017	73.985 ± 0.006	69.673 ± 0.069	71.759 ± 0.038
GSCA [3]	80.687 ± 0.031	74.055 ± 0.073	70.163 ± 0.056	72.964 ± 0.025
TwitterGuard [5]	84.191 ± 0.001	76.153 ± 0.003	73.858 ± 0.010	75.448 ± 0.099
ConvBiLSTM [4]	86.310 ± 0.047	79.078 ± 0.036	77.917 ± 0.079	78.508 ± 0.083
SocioCrimAnalytix [9]	86.965 ± 0.064	81.000 ± 0.061	78.829 ± 0.054	79.250 ± 0.038
CNN-LSTM [79]	89.411 ± 0.021	84.073 ± 0.055	80.255 ± 0.031	82.229 ± 0.066
DAC-BiNet [80]	91.532 ± 0.093	86.817 ± 0.097	85.680 ± 0.048	85.088 ± 0.100
LSA [81]	92.890 ± 0.079	88.473 ± 0.014	87.984 ± 0.076	87.758 ± 0.068
LSTMTwitter [82]	92.315 ± 0.071	89.838 ± 0.092	90.952 ± 0.028	88.268 ± 0.070
MDL [4]	93.674 ± 0.040	90.729 ± 0.094	90.646 ± 0.081	90.125 ± 0.078
Proposed w/o RL	82.325 ± 0.036	76.481 ± 0.030	73.707 ± 0.066	74.662 ± 0.076
Proposed w/o SLF	84.114 ± 0.055	78.317 ± 0.093	75.509 ± 0.019	76.107 ± 0.087
Proposed w/o ABC	88.247 ± 0.036	83.136 ± 0.016	81.918 ± 0.044	82.352 ± 0.036
Proposed	96.411 ± 0.052	93.485 ± 0.021	93.188 ± 0.033	93.944 ± 0.096

TABLE 4. Comparative performance metrics of the proposed model relative to advanced models and ablated studies on the Chicago Twitter (X) data.

Model	Accuracy	Precision	Recall	F-Measure
LSTM [54]	75.296 ± 0.033	69.662 ± 0.034	65.662 ± 0.050	68.829 ± 0.089
GSCA [3]	76.889 ± 0.056	70.933 ± 0.082	66.187 ± 0.010	68.638 ± 0.033
TwitterGuard [5]	80.869 ± 0.003	72.218 ± 0.005	69.534 ± 0.029	71.794 ± 0.088
ConvBiLSTM [4]	82.457 ± 0.002	75.891 ± 0.016	73.921 ± 0.045	74.584 ± 0.063
SocioCrimAnalytix [9]	81.905 ± 0.032	76.631 ± 0.039	74.694 ± 0.034	75.382 ± 0.052
CNN-LSTM [79]	83.666 ± 0.098	79.022 ± 0.026	77.444 ± 0.002	78.118 ± 0.080
DAC-BiNet [80]	87.945 ± 0.098	82.855 ± 0.050	81.069 ± 0.023	81.446 ± 0.008
LSA [81]	89.250 ± 0.039	84.770 ± 0.016	83.987 ± 0.087	83.069 ± 0.003
LSTMTwitter [82]	90.696 ± 0.018	86.403 ± 0.035	85.332 ± 0.088	85.647 ± 0.013
MDL [4]	91.824 ± 0.046	88.768 ± 0.080	87.477 ± 0.038	87.193 ± 0.080
Proposed w/o RL	78.829 ± 0.092	73.993 ± 0.050	70.868 ± 0.043	71.487 ± 0.070
Proposed w/o SLF	79.651 ± 0.054	74.093 ± 0.003	71.078 ± 0.006	72.415 ± 0.051
Proposed w/o ABC	84.480 ± 0.091	80.253 ± 0.030	78.679 ± 0.059	79.997 ± 0.018
Proposed	94.088 ± 0.010	91.434 ± 0.065	91.472 ± 0.078	91.784 ± 0.094

the training process. This close alignment suggests the model generalizes well without overfitting, as evidenced by the minimal divergence between the training and validation performances. Similarly, the loss curves for the Chicago Twitter (X) data also show a steady decrease, with only minor fluctuations between the training and validation losses. This indicates that the model effectively adapts and learns from the new data while maintaining low error rates, thus avoiding overfitting. The slight oscillations in the validation loss towards the later epochs demonstrate the model's capacity to handle real-world data variations effectively. Overall, the consistent performance across both datasets confirms the model's robustness and ability to generalize well across different data types without succumbing to overfitting. This ensures that the model is learning the underlying patterns in the training data and can effectively apply this knowledge to unseen data.

Figure 7 displays the receiver operating characteristic (ROC) curves for the proposed model alongside various advanced models, visually comparing their performance using Chicago crime data and Chicago Twitter (X) data. For Chicago crime data, the analysis indicates that the proposed model significantly outperforms its counterparts with an area under the curve (AUC) of 0.894. This high score suggests that the proposed model effectively distinguishes between

positive and negative outcomes, showcasing a solid ability to predict crime-related occurrences. Notably, models such as MDL and LSA also exhibit commendable performance with AUCs of 0.824 and 0.791, respectively, while LSTM and GSCA display weaker performance with AUCs of 0.613 and 0.644. For Chicago Twitter (X) data, the proposed model maintains the lead with an AUC of 0.903, underscoring its superior predictive accuracy and consistency across different datasets. The model is exceptionally skilled at processing and utilizing social media content to predict outcomes. Other models like MDL and LSTMTwitter also perform well with AUCs of 0.833 and 0.820, respectively, demonstrating their potential in social media analytics. The LSTM model shows a modest improvement in this dataset with an AUC of 0.622, possibly reflecting different feature dynamics in Twitter (X) data compared to crime data. Overall, the comparative analysis in Figure 7 highlights the proposed model's versatility and exceptional performance in utilizing diverse data sources for predictive accuracy in public safety contexts. This capability is particularly beneficial for real-time crime monitoring and proactive policing strategies that leverage traditional crime data and real-time social media insights.

Figure 8 presents the proposed model's precision-recall (PR) curves and several contemporary models, focusing on

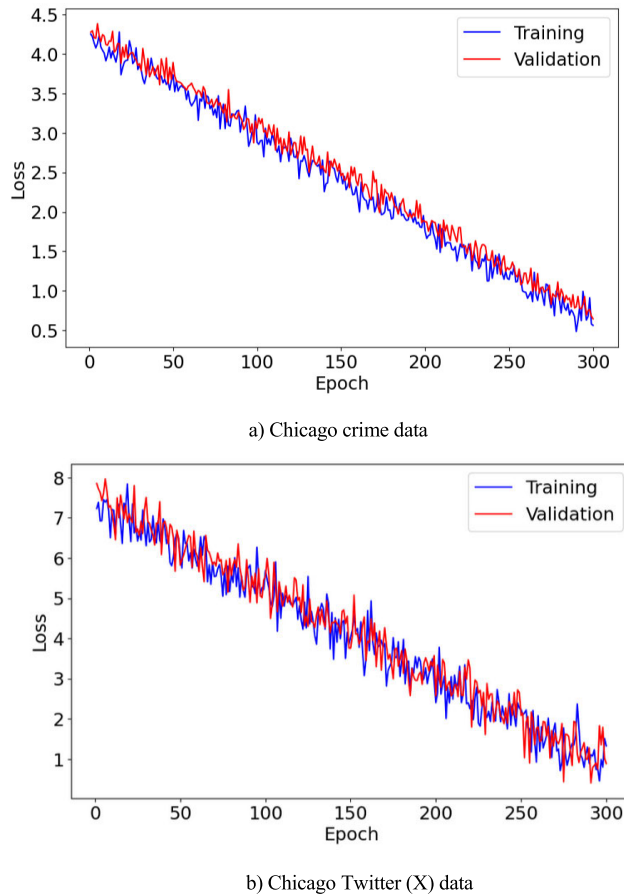


FIGURE 6. Training and validation loss curves in 300 epochs.

the Chicago crime data and Chicago Twitter (X) data. This visualization facilitates a comparison of how each model handles class imbalance. The proposed model distinctly outperforms other models related to Chicago crime data in the PR curve, showcasing a PR-AUC of 0.690. This high score indicates superior management of class imbalances typical in crime data, where positive cases (crimes) may be rare relative to the negative class (non-crimes). The proposed model's curve maintains higher precision at various levels of recall compared to competitors, highlighting its efficacy in identifying relevant patterns without inflating false positives. Such precision is crucial in predictive policing, where accuracy in predictions directly influences resource allocation and public trust. Models like CNN-LSTM and DAC-BiNet also perform well, showing PR-AUCs of 0.614 and 0.629, respectively. These models demonstrate robustness but still need to catch up compared to the proposed model, likely due to less effective handling of the sparse and skewed data distributions in the crime dataset. In contrast, traditional models like LSTM exhibit significantly lower performance, reflecting their limited ability to adjust to the imbalance inherent in crime reporting data. The analysis of the Chicago Twitter (X) data reveals a similar trend, with the proposed model achieving a PR-AUC of 0.733, the highest among the tested models. This

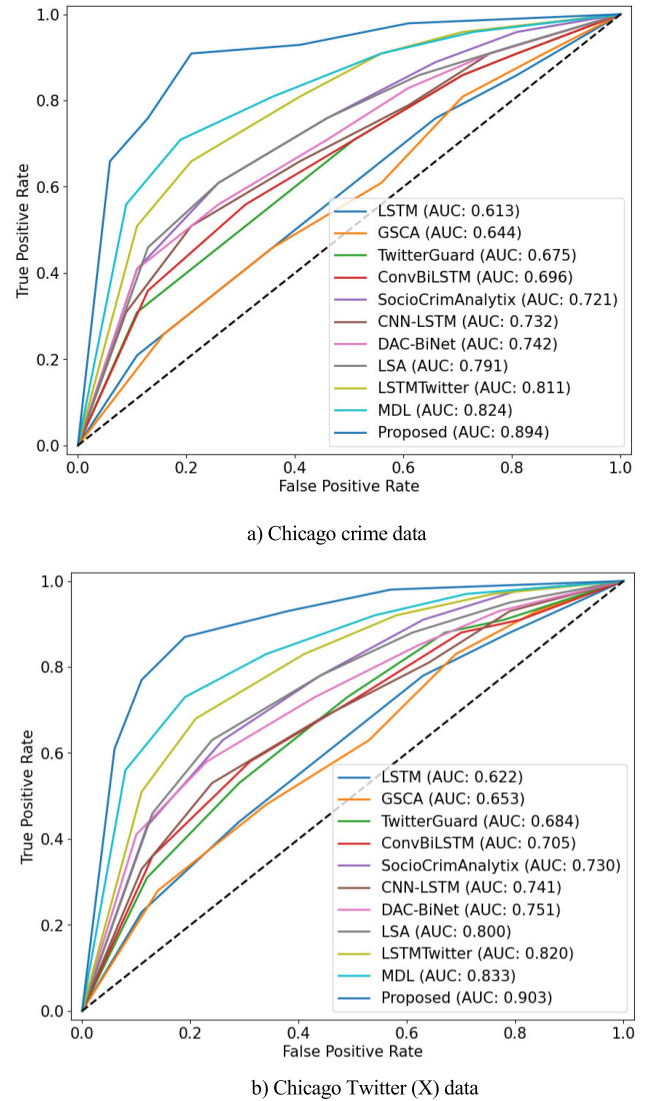


FIGURE 7. Comparative ROC-AUC curve analysis of the proposed model and other models.

result underscores the model's capacity to discern genuinely informative tweets from the most irrelevant posts. High precision across recall levels indicates the model's adeptness at maintaining accuracy even as it attempts to cover more of the positive class, which is critical for applications like sentiment analysis or event detection in noisy social media streams. Other models like MDL and LSTMTwitter show improved performance with PR-AUCs of 0.684 and 0.683, respectively, suggesting that enhancements in model architecture or training strategies are beneficial. However, these must catch up to the proposed model, indicating potential gaps in feature extraction or threshold optimization. The superior performance of the proposed model in both datasets can be attributed to its innovative integration of deep learning architectures and novel algorithmic enhancements. These include dynamic adjustments to the learning process in response to the evolving data landscape, which is particularly effective

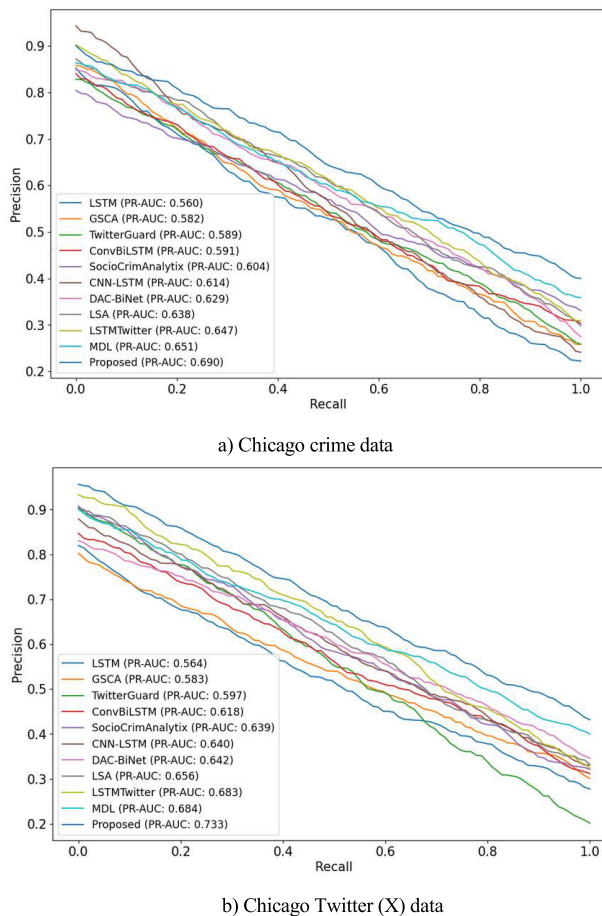


FIGURE 8. Comparative PR-AUC curve analysis of the proposed model and other models.

in scenarios with imbalanced classes. This ability enhances predictive accuracy and ensures the model remains versatile and reliable across diverse and dynamic datasets like those from Chicago's urban environment.

Figure 9 provides a comprehensive visualization of the proposed model's performance scalability across different training data sizes, ranging from 20% to 100%, explicitly applied to the Chicago crime data and Chicago Twitter (X) data. This figure highlights the model's robustness and adaptability to varying amounts of data, a crucial attribute for practical applications in dynamic and data-rich environments. For the Chicago crime dataset, the model showcases a steady increase in all measured metrics as the training size expands. Starting at 20% training data, the accuracy, precision, recall, and F-measure are visibly lower, indicating initial learning stages where the model is yet to capitalize fully on the underlying patterns in the crime data. As the training data size increases to 40% and 60%, there is a notable improvement in precision and recall, which is crucial for crime data analysis, where the accuracy of predicting true crime events significantly enhances the model's utility. When the training data reaches 80% and 100%, the model achieves peak performance metrics, demonstrating its effectiveness

in comprehensively learning from and adapting to the complex features inherent in urban crime data. Similarly, with the Chicago Twitter (X) data, the model starts with modest performance metrics at 20% training data. This initial data provides a basic understanding of public sentiment and crime-related discussions on social media platforms. As more data is included in the training set (40% to 60%), the model's ability to accurately classify and recall relevant tweets improves markedly. This is crucial for applications in social media monitoring, where accurately parsing through vast amounts of data to extract relevant information can provide actionable insights. At 80% and 100% data utilization, the model reaches its optimal performance, illustrating high levels of precision and recall, essential for real-time social media analytics, helping to identify trends, potential threats, or public sentiment accurately. The progressive improvement across all metrics from 20% to 100% training data underscores the proposed model's capability to perform effectively even with limited data. However, its performance is optimized with more extensive data training. This trait is particularly advantageous in real-world scenarios where data availability may vary or in applications requiring rapid deployment with incremental data training. The model's ability to maintain a balanced F-measure throughout the training sizes suggests that it does not sacrifice precision for recall or vice versa. It maintains a robust equilibrium between identifying as many relevant instances as possible (recall) and ensuring they are correct (precision). In conclusion, Figure 9 affirms that the proposed model is well-suited for handling varied and extensive datasets typical of urban settings like Chicago, where crime and social media data present unique analytical challenges. The model's scalability and ability to achieve high performance with increasing amounts of data demonstrate its potential for deployment in diverse applications, ranging from public safety to digital media analysis. The consistent improvement in performance metrics with larger datasets provides confidence in the model's utility in strategic planning and operational responses within these fields.

Figure 10 illustrates the decision-making time distributions for the proposed model when applied to the Chicago crime data and Chicago Twitter (X) data within real-time bidding (RTB) environments. This graph is crucial for understanding the model's efficiency and suitability for applications requiring rapid response times. For the Chicago Crime Data, the decision-making time distribution is predominantly centered around 70-80 milliseconds. This relatively narrow spread indicates the model's consistent and swift response in processing and making decisions on crime-related data. Such quick decision-making is critical in RTB environments where the value of information decreases rapidly with time. The ability to quickly process and react to incoming data ensures that actions can be taken promptly, potentially aiding in crime prevention or resource deployment strategies where seconds can make a difference. In contrast, the decision-making times for the Chicago Twitter (X) Data are slightly faster, with a peak of around 65-75 milliseconds and a broader distribution.

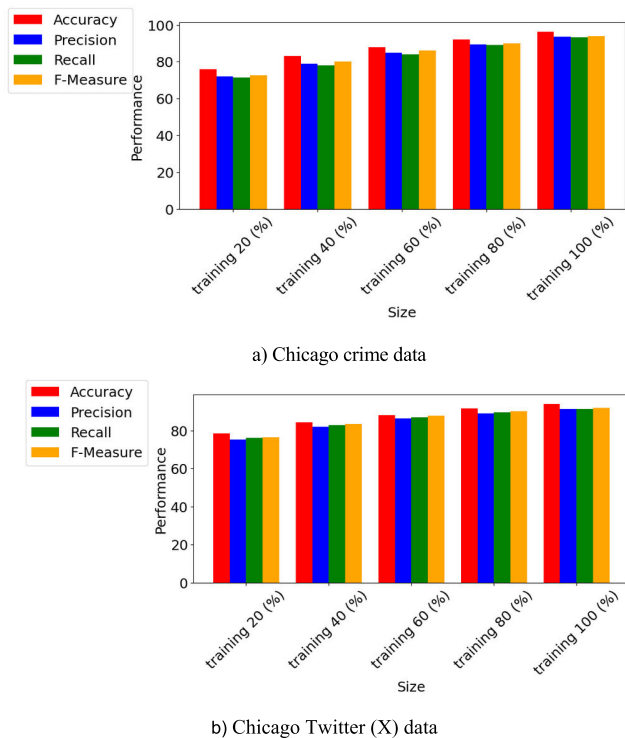


FIGURE 9. Scalability of the proposed model across different training sizes.

This difference may reflect the model's adaptation to the less structured nature of social media data, requiring it to operate even more quickly to filter through more significant volumes of less predictable data. The broader range suggests variability in the complexity of tweets being analyzed, yet the model remains well within the sub-100-millisecond threshold critical for RTB systems. This capability is precious in environments like digital marketing or public sentiment analysis during live events, where real-time data processing can significantly impact decision quality and timing. The efficiency showcased by the model in both datasets confirms its potential for integration into RTB systems where rapid data ingestion and processing are required to make immediate bidding decisions based on real-time analytics.

1) ANALYSIS OF REWARD FUNCTION

In the proposed model, successful identifications of the larger and smaller groups receive rewards of $+1$ and $+\lambda$, respectively, while inaccuracies are penalized by -1 and $-\lambda$. The optimal setting for λ hinges on the ratio of majority to minority samples and generally diminishes as this gap broadens. To assess the effect of varying λ , we methodically increased its values from 0 to 1 in increments of 0.1, and these adjustments are visually represented in Figure 11.

For the Chicago crime data, we observed an enhancement in all performance indicators as λ rose from 0 to approximately 0.3, suggesting that higher rewards for accurately detecting minority samples positively affect the model's

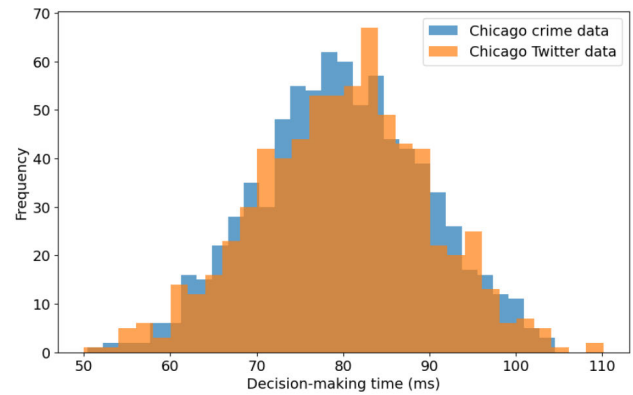


FIGURE 10. Comparison of decision-making time distributions for Chicago crime data and Chicago Twitter (X) data.

overall accuracy. The model achieved its best performance within the λ range of 0.3 to 0.4, indicating this to be the ideal range for the balance parameter in this scenario. However, beyond this optimum, further increases in λ led to a performance downturn, particularly in precision, which plummeted the fastest, hinting at an excessive compensation for the minority class that likely caused more false positives for the majority class. The F-measure, which provides a balanced view of precision and recall, also began to decline but remained relatively stable up to $\lambda = 0.6$ before deteriorating more rapidly, indicating that the model preserved a balance between precision and recall up to a certain threshold before disproportionately favoring the minority class.

In the analysis of tweet data, a similar trend emerged, with performance metrics improving as λ increased up to about 0.4. However, the peak performances for accuracy and the F-measure were more distinct. They occurred at a slightly higher λ value than in the crime data, suggesting that this dataset might require a higher λ value for optimal class balance. Precision and recall showed more fluctuations but generally increased before decreasing alongside λ . Notably, precision began to decline sooner than in the crime data analysis, implying that the model might begin to wrongly classify the majority class as the minority class at lower λ values. After $\lambda = 0.4$, all metrics experienced a significant drop, signaling that the model's effectiveness drastically deteriorates when the reward for correctly identifying the minority class becomes excessively high.

Figure 12 shows the evolving rewards that the agent earns across various episodes, highlighting its learning progression. In the early stages, the agent experiences negative rewards, signaling its initial phase of learning and skill development within the environment. This early phase is crucial as the agent focuses on grasping the essential elements of the task at hand. As the agent gains knowledge and sharpens its analytical abilities, there is a noticeable enhancement in its decision-making capabilities, reflected in the gradual increase of the reward curve. Although the agent may face some setbacks, it demonstrates remarkable resilience,

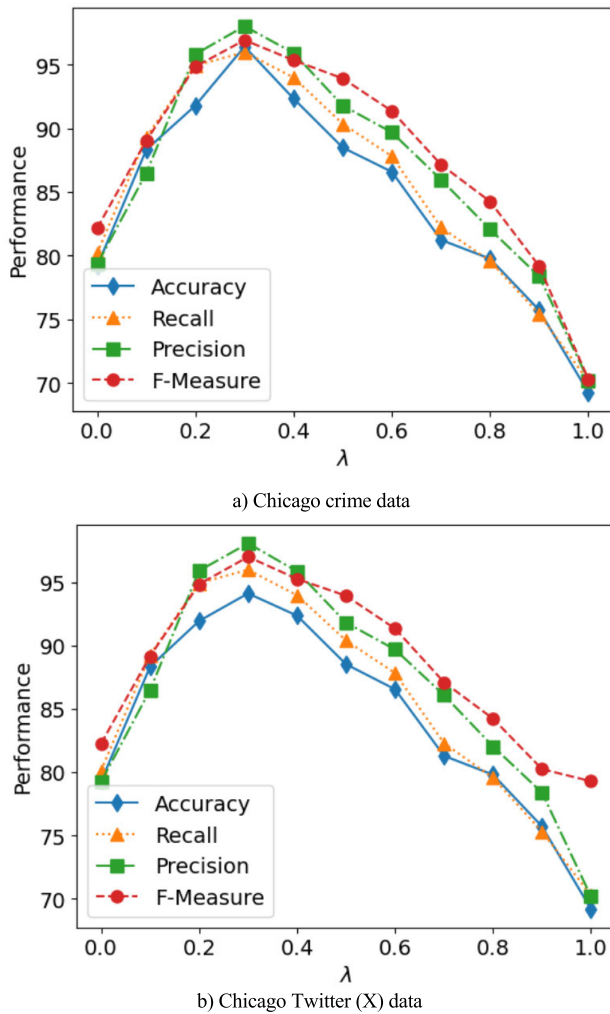


FIGURE 11. Learning progression of the agent through adaptive reward acquisition over episodes.

continually improving its performance. Temporary declines or plateaus in rewards do not hinder the agent's ongoing learning trajectory. The fluctuations in rewards emphasize the agent's dedication to refining its strategies, using past experiences to improve its future decisions. By the end of the training episodes, the agent achieves significant proficiency in navigating its environment, evidenced by the substantial increase in rewards. This improvement highlights the agent's advanced skills and ability to make informed and strategic choices.

2) ANALYZING Q-VALUE DISTRIBUTIONS ACROSS STATES

Figure 13 showcases a series of subplots depicting the Q-values a reinforcement learning agent has associated with ten states, from State 0 to State 9. The agent faces a binary choice in these visual representations: to take action 0 or action 1, as indicated within the plots. The highlighted yellow boxes convey the expected rewards, or Q-values, which quantify the potential benefit of selecting a particular action in a

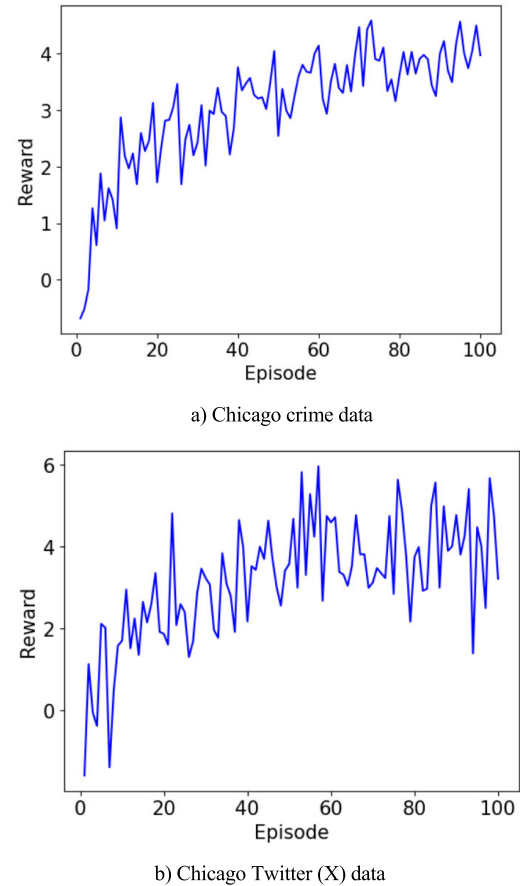
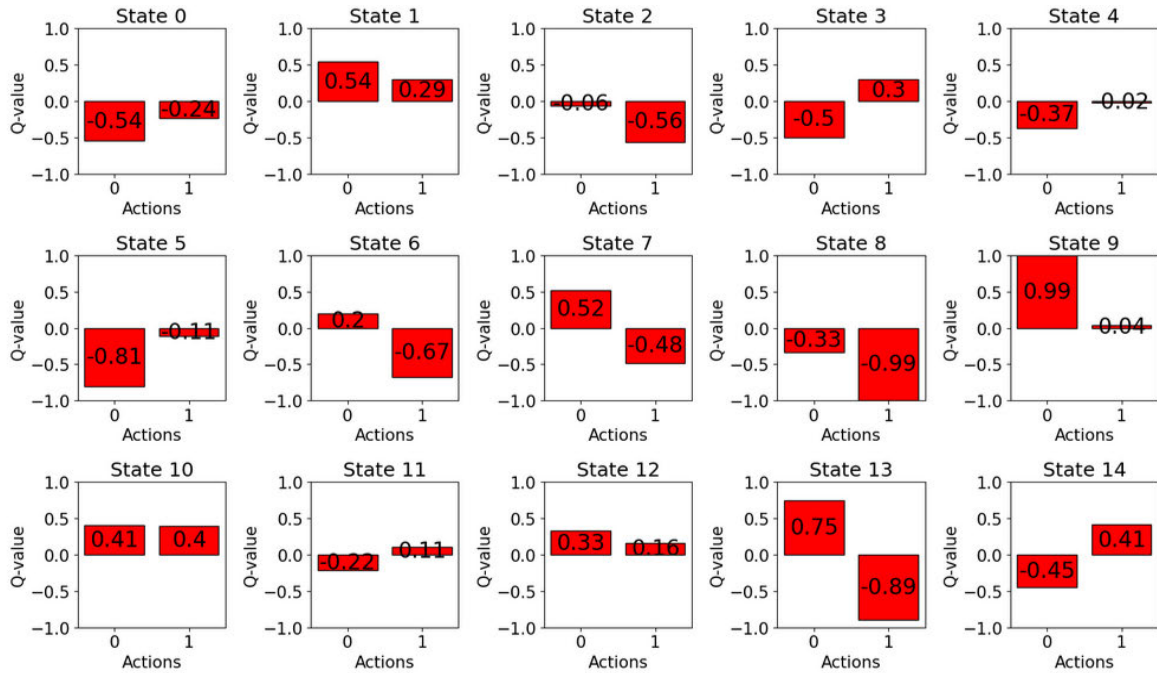


FIGURE 12. Impact of varying reward modifier (λ) on the model performance.

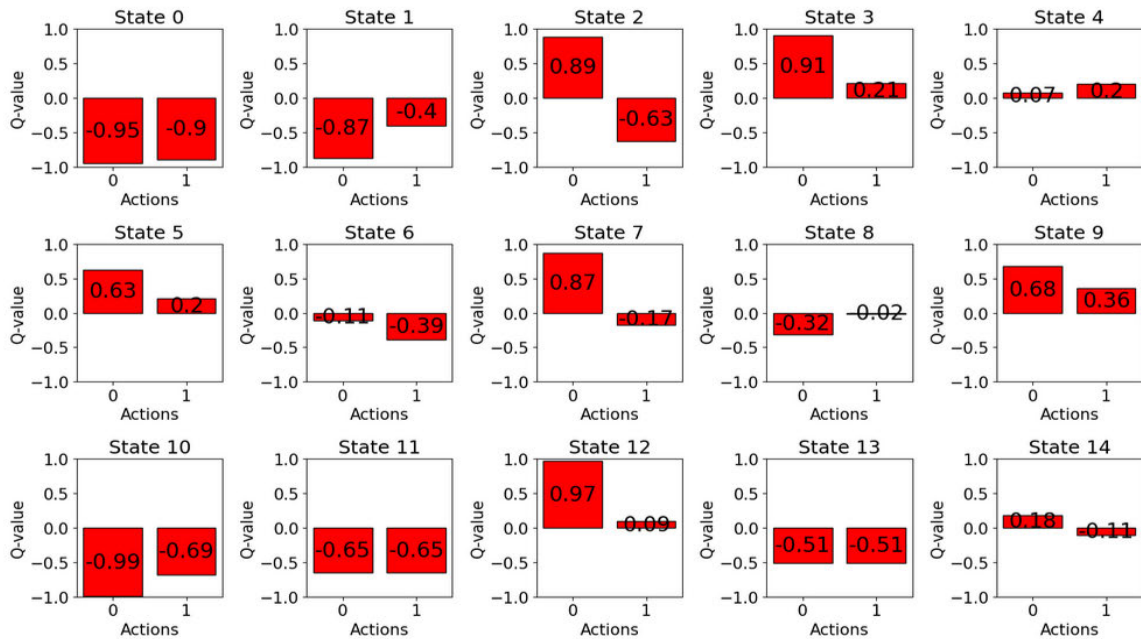
given state. These values are the culmination of the agent's training experience and its interactions with the environment.

Within the crime data context, a pattern emerges from the Q-values that points to the agent's learned preferences. For example, the markedly high Q-value for action 0 in State 9 (0.99) versus a negative value for action 1 (-0.04) signals a clear bias toward action 0, perceived by the agent as more beneficial. Conversely, in State 3, the agent favors action 1, which has a positive Q-value (0.29), over action 0, which is negative (-0.5). This discernible pattern in the Q-values indicates the agent's strategic development, where it has been trained to make choices expected to yield the highest rewards according to the nuances of the crime dataset.

In analyzing tweets, the agent exhibits varied preferences. In State 3, action 0's Q-value is significantly higher (0.91) than action 1 (0.21), suggesting that action 0 is favored in this particular state based on the tweet data. This starkly contrasts the balanced Q-values for both actions in State 10 (-0.7), which implies an absence of preference and an anticipated equal reward from either action. The variation in Q-values, as opposed to those derived from the crime data, underscores the agent's adaptive learning to the distinct characteristics of the tweet dataset, reflecting a nuanced understanding that extends beyond the language to include the context of the



a) Chicago crime data



b) Chicago Twitter (X) data

FIGURE 13. Visualization of Q-Values for ten states in a trained RL agent.

tweets, which is quite different from the structured nature of crime reports.

3) ANALYZE OF WORD EMBEDDING

In this study, we employed DistilBERT as our word embedding method, a cutting-edge approach in the field.

We compared its performance against a range of established embedding techniques, including Skip-gram, CBOW, GloVe, FastText, and the original BERT, as detailed in Tables 5 and 6. DistilBERT demonstrated superior performance, achieving an impressive accuracy of 96.411% on crime data. This marks a substantial improvement over BERT, the next-best model, which earned an accuracy of 93.457%. The approximately

TABLE 5. Comparative performance metrics of DistilBERT relative to other word embeddings on the Chicago crime data.

Model	Accuracy	Precision	Recall	F-Measure
CBOW	81.012 ± 0.083	80.760 ± 0.074	81.198 ± 0.090	80.709 ± 0.009
Skip-gram	82.269 ± 0.043	81.179 ± 0.064	82.645 ± 0.079	81.012 ± 0.066
GloVe	84.963 ± 0.038	84.259 ± 0.043	85.837 ± 0.005	84.140 ± 0.092
FastText	90.117 ± 0.004	88.451 ± 0.079	89.480 ± 0.047	88.713 ± 0.098
BERT	93.185 ± 0.055	91.928 ± 0.037	92.162 ± 0.059	91.113 ± 0.020
DistilBERT	96.411 ± 0.052	93.485 ± 0.021	93.188 ± 0.033	93.944 ± 0.096

TABLE 6. Comparative performance metrics of DistilBERT relative to other word embeddings on Chicago Twitter (X) data.

Model	Accuracy	Precision	Recall	F-Measure
CBOW	80.243 ± 0.017	78.827 ± 0.032	77.046 ± 0.085	77.404 ± 0.029
Skip-gram	79.329 ± 0.057	77.791 ± 0.020	76.495 ± 0.091	76.397 ± 0.057
GloVe	83.096 ± 0.081	81.609 ± 0.066	81.416 ± 0.060	81.361 ± 0.009
FastText	86.739 ± 0.045	85.708 ± 0.082	84.541 ± 0.026	84.990 ± 0.038
BERT	90.115 ± 0.006	88.179 ± 0.031	87.808 ± 0.057	87.565 ± 0.041
DistilBERT	94.088 ± 0.010	91.434 ± 0.065	91.472 ± 0.078	91.784 ± 0.094

3.22% increase in accuracy with DistilBERT highlights its effectiveness in processing complex datasets, such as crime data, characterized by intricate linguistic patterns. Moreover, DistilBERT's precision and recall rates—93.485% and 93.188%, respectively—reflect its exceptional capability to identify relevant data points while minimizing false negatives accurately, a critical factor in the analysis of crime data. The rise in the F-measure to 93.944% with DistilBERT from 91.113% with BERT indicates a harmonious improvement in precision and recall. This equilibrium is crucial for practical applications, ensuring that enhancements in one area do not detract from another. The notable gains across all performance metrics with DistilBERT point to its potential for delivering deeper and more accurate insights in crime data analysis, where precision and reliability are of utmost importance. In evaluating tweet data, DistilBERT again proved its excellence, attaining an accuracy of 94.088%. This represents a significant advancement over BERT, which recorded an accuracy of 91.040%. The roughly 3.048% improvement in accuracy illustrates DistilBERT's proficiency in dealing with the informal and varied language typically found on social media. The precision and recall metrics for DistilBERT, at 91.434% and 91.472%, respectively, denote a high level of accuracy in classifying tweets and effectively reducing the omission of pertinent messages. The F-measure of 91.784% with DistilBERT, compared to 87.026% with BERT, signifies a notable enhancement in the model's ability to balance precision and recall, even within social media's dynamic and unstructured environment. This indicates that DistilBERT is particularly suitable for social media analytics applications, where accurately analyzing and categorizing vast volumes of data can yield significant insights.

4) ANALYZE OF THE PROPOSED ABC ALGORITHM

This study rigorously evaluates the performance of the enhanced ABC algorithm, comparing it against a diverse array of well-known optimization algorithms such as Differential Evolution (DE), Firefly Algorithm (FA), Cuckoo

Optimization Algorithm (COA), Salp Swarm Algorithm (SSA), Bat Algorithm (BA), Human Mental Search (HMS) [83], and the original ABC. The findings in Tables 7 and 8 show how the proposed ABC algorithm compares to these alternatives across two different datasets. On the crime data front, the proposed ABC algorithm stood out with an exceptional accuracy of 96.411%, surpassing the original ABC, its closest rival, which posted an accuracy of 93.457%. This leap in accuracy, approximately 2.954%, highlights the proposed ABC algorithm's proficiency in sifting through complex datasets filled with nuanced patterns, necessitating advanced analytical approaches for precise interpretation. Moreover, the algorithm's precision and recall rates, at 93.485% and 93.188%, respectively, underscore its strong capability to pinpoint relevant instances while effectively and accurately reducing false negatives, a critical consideration in crime data analysis. The F-measure, at 93.944%, further accentuates the algorithm's balanced improvement in precision and recall, reinforcing its reliability and efficiency in extracting valuable insights from crime-related data. Turning to the analysis of tweet data, the proposed ABC algorithm once again delivered standout performance, achieving an accuracy of 93.200%. This marked a significant stride forward from the original ABC, which had an accuracy of 90.462%, representing an improvement rate of about 2.74%. Such results demonstrate the proposed ABC algorithm's skill in navigating social media content's informal and ever-changing landscape, ensuring tweets are classified with remarkable accuracy. The precision and recall figures for the proposed ABC algorithm, at 90.402% and 90.412%, respectively, highlight its effectiveness in categorizing tweets accurately while minimizing overlooking relevant content. The F-measure, at 90.510%, signifies a noteworthy advancement in achieving a balance between precision and recall, an essential aspect in analyzing the unstructured environment of social media.

Figure 14 depicts the trajectory of the optimization process utilizing the proposed ABC algorithm, showing the progression of the objective function values over numerous iterations.

TABLE 7. Comparative performance metrics of the proposed ABC algorithm relative to other optimization algorithms on the Chicago crime data.

Model	Accuracy	Precision	Recall	F-Measure
Proposed+DE	90.982 ± 0.061	88.802 ± 0.009	87.796 ± 0.026	87.957 ± 0.061
Proposed+FA	89.250 ± 0.074	86.012 ± 0.039	85.008 ± 0.029	85.042 ± 0.043
Proposed+COA	80.529 ± 0.027	78.560 ± 0.001	79.760 ± 0.071	78.128 ± 0.091
Proposed+ SSA	86.177 ± 0.070	81.816 ± 0.083	82.439 ± 0.033	81.405 ± 0.055
Proposed+ BA	92.201 ± 0.094	89.481 ± 0.028	90.951 ± 0.051	89.364 ± 0.084
Proposed+HMS	88.354 ± 0.055	85.779 ± 0.011	84.069 ± 0.077	84.354 ± 0.084
Proposed+ABC	93.457 ± 0.053	90.223 ± 0.046	90.431 ± 0.012	90.461 ± 0.035
Proposed ABC	96.411 ± 0.052	93.485 ± 0.021	93.188 ± 0.033	93.944 ± 0.096

TABLE 8. Comparative performance metrics of the proposed ABC algorithm relative to other optimization algorithms on the Chicago Twitter (X) data.

Model	Accuracy	Precision	Recall	F-Measure
Proposed+DE	88.364 ± 0.009	86.518 ± 0.018	85.544 ± 0.009	85.877 ± 0.099
Proposed+FA	86.190 ± 0.036	84.560 ± 0.021	82.926 ± 0.007	83.725 ± 0.093
Proposed+COA	79.750 ± 0.079	78.875 ± 0.068	76.413 ± 0.090	77.182 ± 0.088
Proposed+ SSA	83.977 ± 0.038	82.151 ± 0.027	83.173 ± 0.015	82.926 ± 0.054
Proposed+ BA	89.049 ± 0.100	86.239 ± 0.075	87.895 ± 0.053	86.670 ± 0.000
Proposed+HMS	87.888 ± 0.042	85.063 ± 0.094	85.325 ± 0.033	85.871 ± 0.037
Proposed+ABC	91.040 ± 0.098	87.594 ± 0.046	88.565 ± 0.065	87.026 ± 0.012
Proposed ABC	94.088 ± 0.010	91.434 ± 0.065	91.472 ± 0.078	91.784 ± 0.094

The horizontal axis marks the iteration number, while the vertical axis quantifies the value of the objective function, providing a clear depiction of the algorithm's performance over time. An analysis of Figure 14 reveals significant oscillations in the objective function during the early stages, underscoring the proposed ABC algorithm's initial broad search for feasible solutions. This phase is critical as the algorithm canvasses the solution space extensively, avoiding early settlement on inferior solutions and pinpointing areas with optimal potential. As the iterations advance, these oscillations decline, indicating a transition from broad exploration to targeted exploitation. The algorithm now narrows its focus, honing in on the promising areas identified in the exploratory phase to fine-tune the solution. Monitoring for extended intervals of static objective function values is crucial, as this may signal a stagnation in progress, potentially at a local maximum. Under such circumstances, it may be prudent to modify the proposed ABC algorithm's parameters or to incorporate additional techniques to rejuvenate its search process, thereby steering it toward the global maximum.

5) DISCUSSION

The proposed model in the article presents an innovative approach to forecasting crime using sentiment analysis. It leverages the RL algorithm with the scope loss function and ABC to address the challenges of class imbalance and hyperparameter optimization in sentiment classification. We designed and conducted experiments that demonstrated the superiority of our model in comparison to other advanced models. Furthermore, we carried out ablation studies to highlight the individual contributions of the RL, scope loss, and ABC components to the model's overall performance.

Incorporating RL into our model stems from its ability to learn optimal actions through trial-and-error interactions with a dynamic environment. RL is particularly suited for tasks where the correct input-output mapping is unknown,

making it ideal for sentiment analysis in the context of crime prediction. Using RL, our model learns to make decisions that maximize a reward signal, which, in this case, is aligned with the accurate classification of sentiment classes. This is especially useful in addressing the issue of class imbalance. RL can focus on improving the detection of less frequent minority class instances by adjusting its strategies based on received feedback. This adaptive learning approach enables the model to continuously refine its performance, enhancing accuracy in predicting crime-related sentiments.

Integrating a specialized scope loss function is a strategic choice to tackle the class imbalance problem that is prevalent in sentiment analysis directly. Traditional loss functions like cross-entropy can lead to biased learning towards the majority class, as they do not account for the distribution disparity between classes. The scope loss function, however, is designed to penalize misclassifications of the minority class more heavily, thereby encouraging the model to pay more attention to these critical instances. This approach ensures that the model does not overlook the minority class instances, which are often more informative for crime prediction. By modifying the reward mechanism during the training phase, the scope loss function helps achieve a more balanced and accurate classification across different sentiment classes, improving the overall efficacy of the model in predicting criminal activities from social media content.

The ABC optimization technique was chosen for its effectiveness in fine-tuning hyperparameters, a crucial aspect of deep learning models. The foraging behavior of honey bees inspires ABC and is particularly adept at exploring and exploiting multidimensional search spaces, making it suitable for optimizing the complex landscape of model hyperparameters. This technique helps identify the optimal configuration that yields the best performance, ensuring that the model is accurate and efficient. By employing ABC optimization, we can systematically adjust hyperparameters

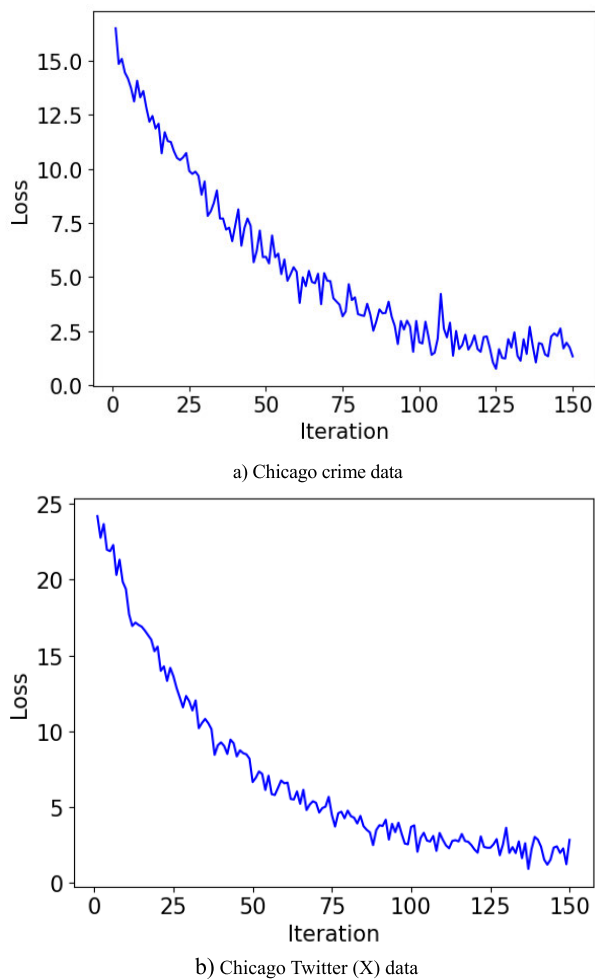


FIGURE 14. Progression of the objective function through consecutive iterations using the proposed ABC algorithm.

such as learning rate, batch size, and network architecture to enhance the model's ability to classify sentiments accurately and predict crime-related activities. The use of ABC contributes to the robustness and adaptability of the model, ensuring high performance across diverse datasets and scenarios.

The theoretical implications of our research are significant. By integrating reinforcement learning with sentiment analysis for crime prediction, this model challenges conventional approaches and introduces a dynamic, adaptive framework for understanding the predictive value of digital communications. This amalgamation underscores the potential of advanced machine learning techniques in extracting actionable insights from vast, unstructured social media data. The model's ability to mitigate class imbalance through a specialized scope loss function and fine-tune its parameters using ABC optimization highlights a novel pathway for addressing longstanding issues in sentiment analysis, such as the accurate classification of minority sentiments. Theoretically, this research extends the applicability of reinforcement learning beyond traditional domains, setting a precedent for its use in social science and criminology. It also contributes to the broader discourse on the ethical and practical implications

of leveraging social media for predictive policing, inviting further exploration into the balance between public safety and privacy.

E. LIMITATIONS

The limitations of the proposed model and its potential solutions are as follows:

- **Dependency on quality and availability of social media data:** A significant limitation of the proposed model is its reliance on the quality and availability of social media data. Social media platforms are dynamic, with constantly changing content, user behavior, and platform policies that can affect data accessibility and representativeness. This variability can introduce biases and inconsistencies in the data, potentially impacting the model's accuracy and generalizability. To address this, future model iterations could incorporate real-time data validation and cleansing mechanisms, ensuring that the input data remains robust and reliable. Additionally, diversifying data sources beyond Twitter (X) to include other platforms could enhance the model's resilience to platform-specific changes and biases.
- **Handling of sarcasm and ambiguity in textual content:** Another limitation is the model's potential difficulty in accurately interpreting sarcasm, irony, and ambiguous language commonly found in social media text. These linguistic nuances can lead to misclassification of sentiments, affecting predictive accuracy. To mitigate this, future enhancements could involve integrating advanced natural language processing techniques, such as contextual embeddings and sarcasm detection algorithms, to improve the model's ability to discern and correctly classify complex linguistic expressions.
- **Scalability and computational efficiency:** The computational complexity of the model, mainly due to the use of reinforcement learning and ABC optimization, poses challenges in terms of scalability and efficiency. As the volume of data and the complexity of the model increase, the computational resources and time required for training and inference can become prohibitive. To overcome this, future work could explore more efficient algorithms and hardware acceleration techniques, such as parallel processing and specialized deep learning hardware, to enhance the model's scalability and reduce computational overhead.
- **Ethical and privacy concerns:** Using social media data for crime prediction raises ethical concerns, particularly regarding user consent, data anonymization, and the potential for biased or discriminatory outcomes. Addressing these concerns is crucial for the responsible deployment of the model. Future developments should include ethical guidelines and privacy-preserving measures, such as differential privacy and data anonymization techniques, to safeguard user information. Moreover, continuous monitoring and auditing mechanisms should be implemented to identify

and mitigate biases, ensuring that the model's predictions do not disproportionately impact any particular group or individual.

VI. CONCLUSION

This paper presented a novel approach that utilized social media sentiment analysis to predict criminal activities. A common challenge in sentiment analysis has been the uneven distribution of sentiment classes, leading conventional models to often misclassify instances of the minority class due to the dominance of majority class data. To tackle this problem, we introduced an RL algorithm combined with a unique scope loss function. This method was designed to reduce class imbalance by altering the reward mechanism during training, improving the precise classification of minority class instances. Our RL-driven training strategy used a series of CNNs to derive features from textual content, which were then integrated for sentiment classification. We also incorporated an advanced ABC optimization technique to adjust the model's hyperparameters precisely. The practical validation of our method used two separate datasets: the first consisted of crime incident reports from the Chicago Police Department, covering September 2019 to July 2024, and the second contained tweets featuring crime-related terms relevant to Chicago. The predictive capacity of our proposed model was compared with other existing models, showing enhanced performance with an accuracy of 96.411% and 94.088%. This notable progress highlights the effectiveness of combining sentiment analysis with reinforcement learning to improve the accuracy of predicting crime-related activities on online social networks, providing valuable insights for law enforcement and criminal investigation efforts.

REFERENCES

- [1] C. Jayapraha, H. S. H. Chitra, and R. M. Priya, "Suspicious crime identification and detection based on social media crime analysis using machine learning algorithms," in *Proc. Soft Comput., Theories Appl. (SoCTA)*. India: Springer, 2023, pp. 831–843.
- [2] S. Aghababaei and M. Makrehchi, "Mining social media content for crime prediction," in *Proc. IEEE/WIC/ACM Int. Conf. Web Intell. (WI)*, Oct. 2016, pp. 526–531.
- [3] Monika and A. Bhat, "Automatic Twitter crime prediction using hybrid wavelet convolutional neural network with world cup optimization," *Int. J. Pattern Recognit. Artif. Intell.*, vol. 36, no. 5, Apr. 2022, Art. no. 2259005.
- [4] S. Tam and Ö. Ö. Tanrıöver, "Multimodal deep learning crime prediction using tweets," *IEEE Access*, vol. 11, pp. 93204–93214, 2023.
- [5] M. Boukabous and M. Azizi, "Crime prediction using a hybrid sentiment analysis approach based on the bidirectional encoder representations from transformers," *Indonesian J. Electr. Eng. Comput. Sci.*, vol. 25, no. 2, p. 1131, Feb. 2022.
- [6] R. Mahajan and V. Mansotra, "Correlating crime and social media: Using semantic sentiment analysis," *Int. J. Adv. Comput. Sci. Appl.*, vol. 12, no. 3, pp. 309–316, 2021.
- [7] M. Hasan, M. T. Rahman, A. H. Zillane, M. G. R. Alam, M. F. Ul Islam, and A. Chakrabarty, "Multilingual sentiment analysis on social media: Harnessing deep learning for enhanced insights and decision support for foreign travelers," in *Proc. 6th Int. Conf. Electr. Eng. Inf. Commun. Technol. (ICEEICT)*, vol. 12, May 2024, pp. 166–171.
- [8] M. Wahiduzzaman, M. Fahim-Ul-Islam, and A. Chakrabarty, "Decoding emotions: Leveraging machine learning to analyze emotional expression for deeper mental health insights," in *Proc. Int. Conf. Adv. Comput., Commun., Electr., Smart Syst. (iACCESS)*, 2024, pp. 1–6.
- [9] M. Vivek and B. R. Prathap, "Spatio-temporal crime analysis and forecasting on Twitter data using machine learning algorithms," *Social Netw. Comput. Sci.*, vol. 4, no. 4, p. 383, May 2023.
- [10] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *J. Artif. Intell. Res.*, vol. 16, pp. 321–357, Jun. 2002.
- [11] S. Zhou, Y. Gu, H. Yu, X. Yang, and S. Gao, "RUE: A robust personalized cost assignment strategy for class imbalance cost-sensitive learning," *J. King Saud Univ. Comput. Inf. Sci.*, vol. 35, no. 4, pp. 36–49, Apr. 2023.
- [12] F. R. Adi Pratama and S. I. Oktora, "Synthetic minority over-sampling technique (SMOTE) for handling imbalanced data in poverty classification," *Stat. J. IAOS*, vol. 39, no. 1, pp. 233–239, Mar. 2023.
- [13] O. Volk and G. Singer, "An adaptive cost-sensitive learning approach in neural networks to minimize local training-test class distributions mismatch," *Intell. Syst. Appl.*, vol. 21, Mar. 2024, Art. no. 200316.
- [14] S. Fu, Y. Tian, J. Tang, and X. Liu, "Cost-sensitive learning with modified stein loss function," *Neurocomputing*, vol. 525, pp. 57–75, Mar. 2023.
- [15] R. Almarshdi, L. Nassef, E. Fadel, and N. Alowidi, "Hybrid deep learning based attack detection for imbalanced data classification," *Intell. Autom. Soft Comput.*, vol. 35, no. 1, pp. 297–320, 2023.
- [16] H. Ding, Y. Sun, Z. Wang, N. Huang, Z. Shen, and X. Cui, "RGAN-EL: A GAN and ensemble learning-based hybrid approach for imbalanced data classification," *Inf. Process. Manage.*, vol. 60, no. 2, Mar. 2023, Art. no. 103235.
- [17] D. Roy, R. Pramanik, and R. Sarkar, "Margin-aware adaptive-weighted-loss for deep learning based imbalanced data classification," *IEEE Trans. Artif. Intell.*, vol. 5, no. 2, pp. 1–10, Feb. 2023.
- [18] M. Esmaeili, S. H. Goki, B. H. K. Masjidi, M. Sameh, H. Gharagozlou, and A. S. Mohammed, "ML-DDoSnet: IoT intrusion detection based on denial-of-service attacks using machine learning methods and NSL-KDD," *Wireless Commun. Mobile Comput.*, vol. 2022, pp. 1–16, Aug. 2022.
- [19] S. V. Moravvej, R. Alizadehsani, S. Khanam, Z. Sobhaninia, A. Shoeibi, F. Khozimeh, Z. A. Sani, R.-S. Tan, A. Khosravi, S. Nahavandi, N. A. Kadri, M. M. Azizan, N. Arunkumar, and U. R. Acharya, "RLMD-PA: A reinforcement learning-based myocarditis diagnosis combined with a population-based algorithm for pretraining weights," *Contrast Media Mol. Imag.*, vol. 2022, no. 1, pp. 1–15, Jan. 2022.
- [20] P. Ebrahimi, H. Dustmohammadloo, H. Kabiri, P. Bouzari, and M. Fekete-Farkas, "Transformational entrepreneurship and digital platforms: A combination of ISM-MICMAC and unsupervised machine learning algorithms," *Big Data Cognit. Comput.*, vol. 7, no. 2, p. 118, Jun. 2023.
- [21] S. Danaei, A. Bostani, S. V. Moravvej, F. Mohammadi, R. Alizadehsani, A. Shoeibi, H. Alinejad-Rokny, and S. Nahavandi, "Myocarditis diagnosis: A method using mutual learning-based ABC and reinforcement learning," in *Proc. IEEE 22nd Int. Symp. Comput. Intell. Informat. 8th IEEE Int. Conf. Recent Achievements Mechatronics, Autom., Comput. Sci. Robot. (CINTI-MACRO)*, Nov. 2022, pp. 000265–000270.
- [22] H. Dustmohammadloo, A. Nikeghbalzadeh, F. F. Karkaragh, and S. Souri, "Knowledge sharing as a moderator between organizational learning and error management culture in Academic staff," *Int. J. Multicultural Educ. (IJME)*, vol. 25, no. 3, pp. 56–73, 2023.
- [23] M. Roshanzamir, R. Alizadehsani, S. V. Moravvej, J. H. Joloudari, H. Alinejad-Rokny, and J. M. Gorriz, "Enhancing interpretability in machine learning: A focus on genetic network programming, its variants, and applications," in *Proc. Int. Work-Conf. Interplay Between Natural Artif. Comput. Portugal: Springer*, 2024, pp. 98–107.
- [24] Z. Ghasemi Kooktapeh, H. Dustmohammadloo, H. Mehrdoost, and F. Fatehi, "In the line of fire: A systematic review and meta-analysis of job burnout among nurses," 2023, *arXiv:2312.14853*.
- [25] S. Taherinavid, S. V. Moravvej, Y.-L. Chen, J. Yang, C. S. Ku, and P. L. Yee, "Automatic transportation mode classification using a deep reinforcement learning approach with smartphone sensors," *IEEE Access*, vol. 12, pp. 514–533, 2024.
- [26] J. Hao, T. Yang, H. Tang, C. Bai, J. Liu, Z. Meng, P. Liu, and Z. Wang, "Exploration in deep reinforcement learning: From single-agent to multi-agent domain," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 7, pp. 1–21, Jul. 2023.
- [27] M. H. Modirrousta, M. Aliyari Shoorehdeli, M. Yari, and A. Ghahremani, "Deep Q-learning recommender algorithm with update policy for a real steam turbine system," *IET Collaborative Intell. Manuf.*, vol. 5, no. 3, pp. 1–11, Sep. 2023.

- [28] P. Saeid, M. Zeinolabedini, and M. Khamforoush, "Simulation of a crossflow ultrafiltration polysulfone/polyvinylpyrrolidone membrane separation using finite element analysis to separate oil/water emulsion," *Iranian Polym. J.*, vol. 32, no. 4, pp. 447–455, Apr. 2023.
- [29] M. Li, X. Zhang, C. Thrampoulidis, J. Chen, and S. Oymak, "Autobalance: Optimized loss functions for imbalanced data," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 3163–3177.
- [30] E. Lin, Q. Chen, and X. Qi, "Deep reinforcement learning for imbalanced classification," *Appl. Intell.*, vol. 50, pp. 2488–2502, Aug. 2020.
- [31] V. Mnih, K. Kavukcuoglu, D. Silver, A. Graves, I. Antonoglou, D. Wierstra, and M. Riedmiller, "Playing Atari with deep reinforcement learning," 2013, *arXiv:1312.5602*.
- [32] V. Mnih, A. P. Badia, M. Mirza, A. Graves, T. Lillicrap, T. Harley, D. Silver, and K. Kavukcuoglu, "Asynchronous methods for deep reinforcement learning," in *Proc. Int. Conf. Mach. Learn.*, 2016, pp. 1928–1937.
- [33] D. Zeng, T. Yan, Z. Zeng, H. Liu, and P. Guan, "A hyperparameter adaptive genetic algorithm based on DQN," *J. Circuits, Syst. Comput.*, vol. 32, no. 4, Mar. 2023, Art. no. 2350062.
- [34] A. Kokhahi and D. Li, "GLHAD: A group lasso-based hybrid attack detection and localization framework for multistage manufacturing systems," *J. Comput. Inf. Sci. Eng.*, vol. 24, no. 7, pp. 1–10, Jul. 2024.
- [35] Y. Sun, Y. Chen, P. Wu, X. Wang, and Q. Wang, "DRL: Dynamic rebalance learning for adversarial robustness of UAV with long-tailed distribution," *Comput. Commun.*, vol. 205, pp. 14–23, May 2023.
- [36] Y. Ali, E. Awwad, M. Al-Razgan, and A. Maarouf, "Hyperparameter search for machine learning algorithms for optimizing the computational complexity," *Processes*, vol. 11, no. 2, p. 349, Jan. 2023.
- [37] C. Erden, "Genetic algorithm-based hyperparameter optimization of deep learning models for PM2.5 time-series prediction," *Int. J. Environ. Sci. Technol.*, vol. 20, no. 3, pp. 2959–2982, Mar. 2023.
- [38] S. Sah, B. Surendiran, R. Dhanalakshmi, and M. Yamin, "Covid-19 cases prediction using SARIMAX model by tuning hyperparameter through grid search cross-validation approach," *Expert Syst.*, vol. 40, no. 5, Jun. 2023, Art. no. e13086.
- [39] M. Daviran, M. Shamekhi, R. Ghezelbash, and A. Maghsoudi, "Land-slide susceptibility prediction using artificial neural networks, SVMs and random forest: Hyperparameters tuning by genetic optimization algorithm," *Int. J. Environ. Sci. Technol.*, vol. 20, no. 1, pp. 259–276, Jan. 2023.
- [40] B. Mohan and J. Badra, "A novel automated SuperLearner using a genetic algorithm-based hyperparameter optimization," *Adv. Eng. Softw.*, vol. 175, Jan. 2023, Art. no. 103358.
- [41] D. Karaboga and B. Basturk, "A powerful and efficient algorithm for numerical function optimization: Artificial bee colony (ABC) algorithm," *J. Global Optim.*, vol. 39, no. 3, pp. 459–471, Oct. 2007.
- [42] A. Karaman, D. Karaboga, I. Pacal, B. Akay, A. Basturk, U. Nalbantoglu, S. Coskun, and O. Sahin, "Hyper-parameter optimization of deep learning architectures using artificial bee colony (ABC) algorithm for high performance real-time automatic colorectal cancer (CRC) polyp detection," *Appl. Intelligence*, vol. 53, no. 12, pp. 15603–15620, 2023.
- [43] U. Erkan, A. Toktas, and D. Ustun, "Hyperparameter optimization of deep CNN classifier for plant species identification using artificial bee colony algorithm," *J. Ambient Intell. Humanized Comput.*, vol. 14, no. 7, pp. 8827–8838, Jul. 2023.
- [44] A. A. Alsubayhin, B. Alzahrani, and M. S. Ramzan, "Review of crime prediction through machine learning," *Int. J. Intell. Syst. Appl. Eng.*, vol. 12, no. 5s, pp. 273–281, 2024.
- [45] R. Ganesan and S. Ravichandran, "Performance analysis for crime prediction and detection using machine learning algorithms," *Int. J. Intell. Syst. Appl. Eng.*, vol. 12, no. 14s, pp. 348–355, 2024.
- [46] K. Ranaweera, "The geography of female fear of crime: A bibliometric mapping and an overview in an Asian context," *Geography*, vol. 1, no. 2, pp. 1–27, Nov. 2024.
- [47] D. Irawan, D. I. Sensuse, P. A. W. Putro, and A. Prasetyo, "Public response to the legalization of the criminal code bill with Twitter data sentiment analysis," *Int. J. Adv. Comput. Sci. Appl.*, vol. 14, no. 2, pp. 295–303, 2023.
- [48] T. M. Oladele and E. F. Ayetiran, "Social unrest prediction through sentiment analysis on Twitter using support vector machine: Experimental study on Nigeria's #EndSARS," *Open Inf. Sci.*, vol. 7, no. 1, Mar. 2023, Art. no. 20220141.
- [49] M. Saraiva, I. Matijošaitienė, S. Mishra, and A. Amante, "Crime prediction and monitoring in Porto, Portugal, using machine learning, spatial and text analytics," *ISPRS Int. J. Geo-Inf.*, vol. 11, no. 7, p. 400, Jul. 2022.
- [50] Z. Abbass, Z. Ali, M. Ali, B. Akbar, and A. Saleem, "A framework to predict social crime through Twitter tweets by using machine learning," in *Proc. IEEE 14th Int. Conf. Semantic Comput. (ICSC)*, Feb. 2020, pp. 363–368.
- [51] T. Vo, R. Sharma, R. Kumar, L. H. Son, B. T. Pham, D. T. Bui, I. Priyadarshini, M. Sarkar, and T. Le, "Crime rate detection using social media of different crime locations and Twitter part-of-speech tagger with Brown clustering," *J. Intell. Fuzzy Syst.*, vol. 38, no. 4, pp. 4287–4299, Apr. 2020.
- [52] A. Ristea, M. Al Boni, B. Resch, M. S. Gerber, and M. Leitner, "Spatial crime distribution and prediction for sporting events using social media," *Int. J. Geographical Inf. Sci.*, vol. 34, no. 9, pp. 1708–1739, Sep. 2020.
- [53] V. Mahor, R. Rawat, S. Telang, B. Garg, D. Mukhopadhyay, and P. Palimkar, "Machine learning based detection of cyber crime hub analysis using Twitter data," in *Proc. IEEE 4th Int. Conf. Comput., Power Commun. Technol. (GUCON)*, Sep. 2021, pp. 1–5.
- [54] C. Sandagiri, B. T. G. S. Kumara, and B. Kuhaneswaran, "Deep neural network-based crime prediction using Twitter data," *Int. J. Syst. Service-Oriented Eng.*, vol. 11, no. 1, pp. 15–30, Jan. 2021.
- [55] S. P. C. W. Sandagiri, B. T. G. S. Kumara, and B. Kuhaneswaran, "ANN based crime detection and prediction using Twitter posts and weather data," in *Proc. Int. Conf. Data Analytics Bus. Industry, Way Towards a Sustain. Economy (ICDABI)*, Oct. 2020, pp. 1–5.
- [56] G. Deepak, S. Rooban, and A. Santhanavijayan, "A knowledge centric hybridized approach for crime classification incorporating deep bi-LSTM neural network," *Multimedia Tools Appl.*, vol. 80, no. 18, pp. 28061–28085, Jul. 2021.
- [57] C. Colas, O. Sigaud, and P.-Y. Oudeyer, "GEP-PG: Decoupling exploration and exploitation in deep reinforcement learning algorithms," in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 1039–1048.
- [58] M. Simchowit and A. Slivkins, "Exploration and incentives in reinforcement learning," *Oper. Res.*, vol. 72, no. 3, p. 495, 2023.
- [59] V. Padhye and K. Lakshmanan, "A deep actor critic reinforcement learning framework for learning to rank," *Neurocomputing*, vol. 547, Aug. 2023, Art. no. 126314.
- [60] Y. Jia and X. Y. Zhou, "Policy gradient and actor-critic learning in continuous time and space: Theory and algorithms," *J. Mach. Learn. Res.*, vol. 23, no. 1, pp. 12603–12652, 2022.
- [61] J. Du, J. Futoma, and F. Doshi-Velez, "Model-based reinforcement learning for semi-Markov decision processes with neural odes," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 19805–19816.
- [62] H. Yang and X. Xie, "An actor-critic deep reinforcement learning approach for transmission scheduling in cognitive Internet of Things systems," *IEEE Syst. J.*, vol. 14, no. 1, pp. 51–60, Mar. 2020.
- [63] P. Saeid, M. Pazoki, and M. Zeinolabedini, "Optimization of biomass production from sugar bagasse in anaerobic digestion using genetic algorithm," *Model. Earth Syst. Environ.*, vol. 9, no. 2, pp. 2183–2198, Jun. 2023, doi: [10.1007/s40808-022-01598-x](https://doi.org/10.1007/s40808-022-01598-x).
- [64] L. Li, D. Li, T. Song, and X. Xu, "Actor-critic learning control with regularization and feature selection in policy gradient estimation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 3, pp. 1217–1227, Mar. 2021.
- [65] S. Vakilian, S. V. Moravvej, and A. Fanian, "Using the artificial bee colony (ABC) algorithm in collaboration with the fog nodes in the Internet of Things three-layer architecture," in *Proc. 29th Iranian Conf. Electr. Eng. (ICEE)*, May 2021, pp. 509–513.
- [66] H. Gharagozlou, J. Mohammadzadeh, A. Bastanfard, and S. S. Ghidary, "RLAS-BIABC: A reinforcement learning-based answer selection using the BERT model boosted by an improved ABC algorithm," *Comput. Intell. Neurosci.*, vol. 2022, pp. 1–21, May 2022, doi: [10.1155/2022/7839840](https://doi.org/10.1155/2022/7839840).
- [67] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter," 2019, *arXiv:1910.01108*.
- [68] A. Y. Vaziri, B. Makkiabadi, and N. Samadzadehaghdam, "EEGg: Generating synthetic EEG signals in MATLAB environment," *Frontiers Biomed. Technol.*, vol. 10, no. 3, pp. 370–381, 2023.
- [69] M. Attari, N. P. Nguyen, K. Palaniappan, and F. Bunyak, "Multi-loss topology-aware deep learning network for segmentation of vessels in microscopy images," in *Proc. IEEE Appl. Imag. Pattern Recognit. Workshop (AIPR)*, vol. 32, Sep. 2023, pp. 1–7.
- [70] P. R. Kotha, M. Attari, M. Maschmann, and F. Bunyak, "Deep style transfer for generation of photo-realistic synthetic images of CNT forests," in *Proc. IEEE Appl. Imag. Pattern Recognit. Workshop (AIPR)*, vol. 30, Sep. 2023, pp. 1–7.

- [71] M. L. Alarcon, U. Roy, A. Kambhampati, N. Nguyen, M. Attari, R. Surya, F. Bunyak, M. Maschmann, K. Palaniappan, and P. Calyam, "Learning-based image analytics in user-AI agent interactions for cyber-enabled manufacturing," in *Proc. IEEE 4th Int. Conf. Human-Machine Syst. (ICHMS)*, vol. 22, May 2024, pp. 1–7.
- [72] J. Han, C. Gondro, K. Reid, and J. P. Steibel, "Heuristic hyperparameter optimization of deep learning models for genomic prediction," *G3 Genes[Genomes]Genetics*, vol. 11, no. 7, Jul. 2021, Art. no. jkab032.
- [73] J. C. Bean, "Genetic algorithms and random keys for sequencing and optimization," *ORSA J. Comput.*, vol. 6, no. 2, pp. 154–160, May 1994.
- [74] *Chicago Data Web Portal*. Accessed: Aug. 9, 2024. [Online]. Available: <https://data.cityofchicago.org/>
- [75] A. M. M. Kasmaee, A. Ataei, S. V. Moravvej, R. Alizadehsani, J. M. Gorriz, Y.-D. Zhang, R.-S. Tan, and U. R. Acharya, "ELRL-MD: A deep learning approach for myocarditis diagnosis using cardiac magnetic resonance images with ensemble and reinforcement learning integration," *Physiological Meas.*, vol. 45, no. 5, May 2024, Art. no. 055011.
- [76] N. D. Arianti, E. Saputra, and A. Sitorus, "An automatic generation of pre-processing strategy combined with machine learning multivariate analysis for NIR spectral data," *J. Agricult. Food Res.*, vol. 13, Sep. 2023, Art. no. 100625.
- [77] R. Teimas and J. Saia, "Detecting persuasion attempts on social networks: Unearthing the potential of loss functions and text pre-processing in imbalanced data settings," *Electronics*, vol. 12, no. 21, p. 4447, Oct. 2023.
- [78] L. A. Yates, Z. Aandahl, S. A. Richards, and B. W. Brook, "Cross validation for model selection: A review with examples from ecology," *Ecological Monographs*, vol. 93, no. 1, p. e1557, Feb. 2023.
- [79] Z. K. Abdalrda, A. M. Al-Bakry, and A. K. Farhan, "A hybrid CNN-LSTM and XGBoost approach for crime detection in tweets using an intelligent dictionary," *Revue d'Intelligence Artificielle*, vol. 37, no. 6, pp. 1651–1661, Dec. 2023.
- [80] Monika and A. Bhat, "DAC-BiNet: Twitter crime detection using deep attention convolutional bi-directional Aquila optimal network," *Multimedia Tools Appl.*, vol. 83, no. 15, pp. 44121–44145, Oct. 2023.
- [81] D. J. W. Wise, S. Ambareesh, P. R. Babu, D. Sugumar, J. P. Bhimavarapu, and A. S. Kumar, "Latent semantic analysis based sentimental analysis of tweets in social media for the classification of cyberbullying text," *Int. J. Intell. Syst. Appl. Eng.*, vol. 12, no. 7s, pp. 26–35, 2024.
- [82] P. Kaladevi, M. Gopalraj, K. Harish, A. Guna, and R. Praveen, "Tweets to predict: LSTM model for crime analysis using Twitter time series data," in *Proc. Int. Conf. Knowl. Eng. Commun. Syst. (ICKECS)*, Apr. 2024, pp. 1–6.
- [83] S. V. Moravvej, S. J. Mousavirad, D. Oliva, G. Schaefer, and Z. Sobhaninia, "An improved DE algorithm to optimise the learning process of a BERT-based plagiarism detection model," in *Proc. IEEE Congr. Evol. Comput. (CEC)*, Jul. 2022, pp. 1–7.



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